



OSCILLATIONS, WAVES AND OPTICS

PHYSICS FOR CHEMISTRY STUDENTS

Ye Yang
Xiamen University

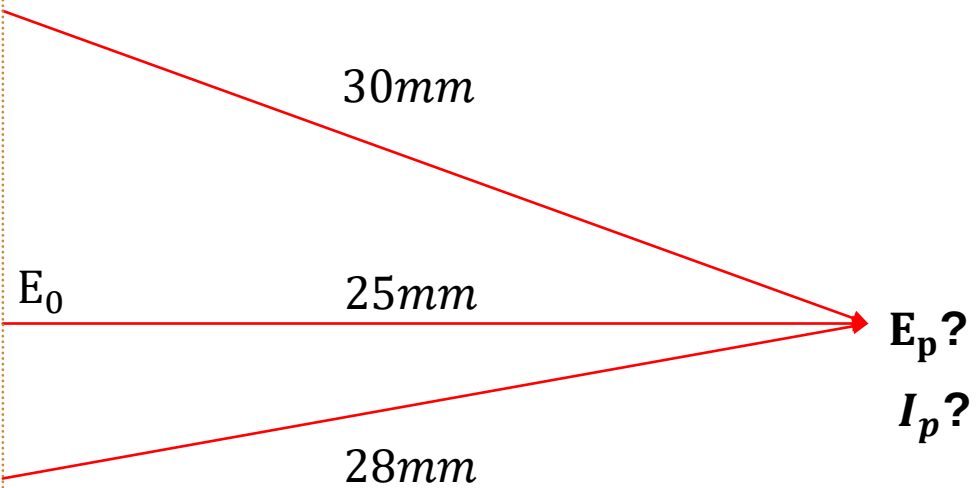
QUIZ

$$\lambda = 600nm$$

$$E_1(r, t) = E_0 \sin(kz + \omega t + \varphi_0)$$

$$E_2(r, t) = E_0 \sin(kz + \omega t + \varphi_0)$$

$$E_3(r, t) = E_0 \sin(kz + \omega t + \varphi_0)$$



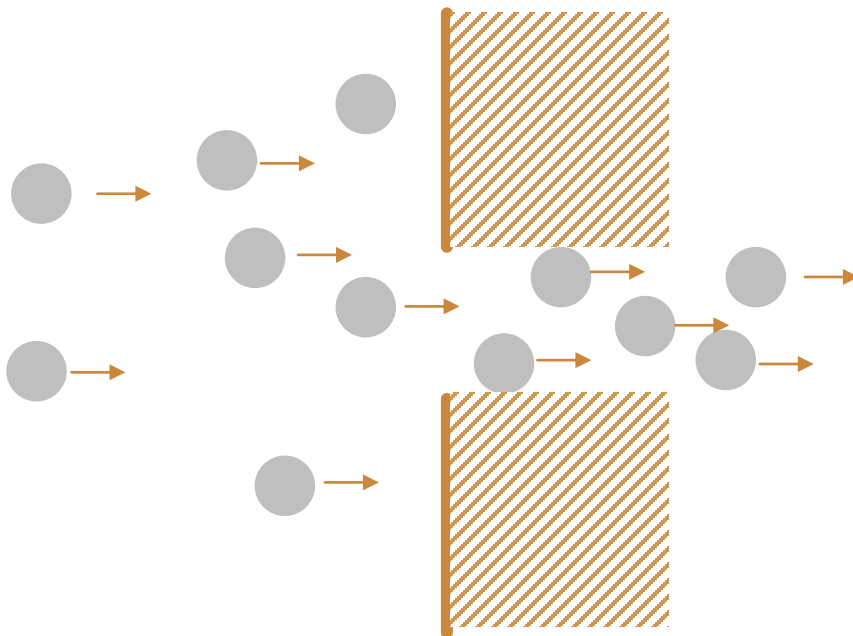
ANSWER

$$E_{00}^2 = \left(\sum_{i=1}^n E_{0i} \sin \alpha_{0i} \right)^2 + \left(\sum_{i=1}^n E_{0i} \cos \alpha_{0i} \right)^2 \quad 4.2 \text{ Superposition}$$

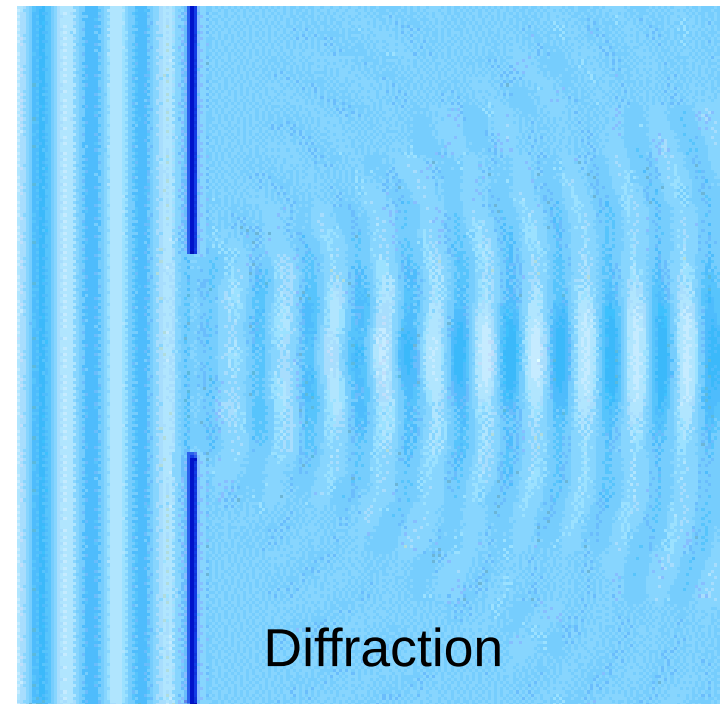
$$\begin{aligned} &= E_{01}^2 + E_{02}^2 + E_{03}^2 + 2E_{01}E_{02}(\cos \alpha_{01} \cos \alpha_{02} + \sin \alpha_{01} \sin \alpha_{02}) \\ &+ 2E_{01}E_{03}(\cos \alpha_{01} \cos \alpha_{03} + \sin \alpha_{01} \sin \alpha_{03}) \\ &+ 2E_{03}E_{02}(\cos \alpha_{03} \cos \alpha_{02} + \sin \alpha_{03} \sin \alpha_{02}) \\ &= E_{01}^2 + E_{02}^2 + E_{03}^2 + 2E_{01}E_{02} \cos(\alpha_{01} - \alpha_{02}) + 2E_{01}E_{03} \cos(\alpha_{01} - \alpha_{03}) \\ &+ 2E_{03}E_{02} \cos(\alpha_{03} - \alpha_{02}) \end{aligned}$$

4.6 FRAUNHOFER DIFFRACTION

Particles

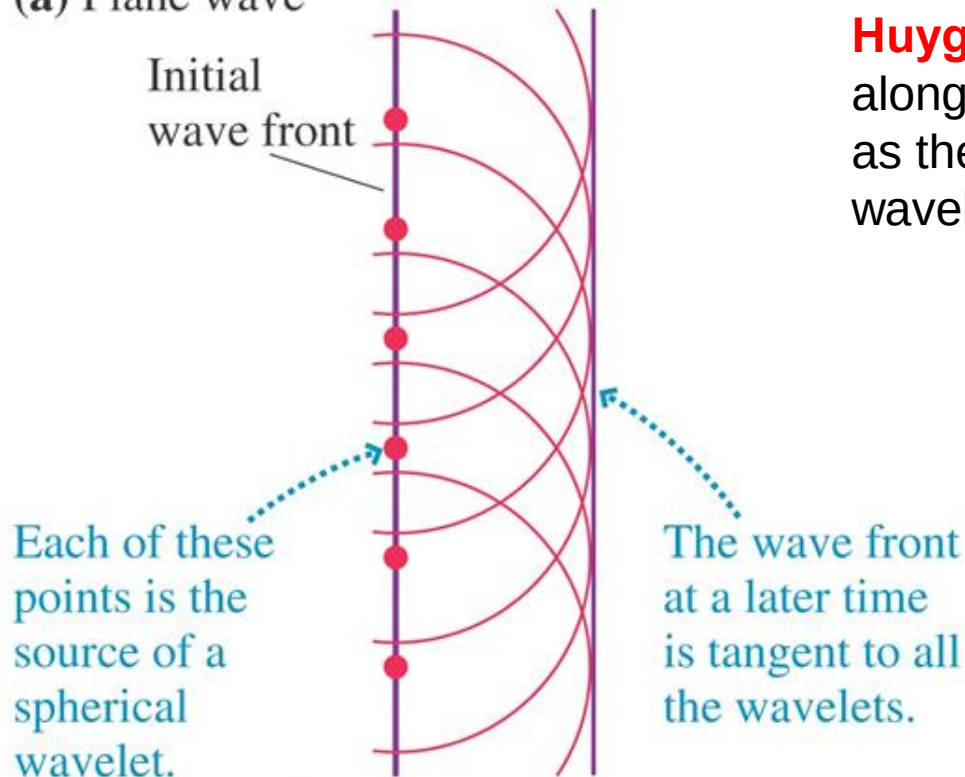


waves



4.6 FRAUNHOFER DIFFRACTION

(a) Plane wave



Huygens' principle: every point along a wavefront can be treated as the source of a spherical wavelet

4.6 FRAUNHOFER DIFFRACTION

$$\frac{\partial^2}{\partial t^2} \mathbf{E}(r, t) = \frac{1}{\epsilon_0 \mu_0} \nabla^2 \mathbf{E}(r, t)$$

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}$$

Spherical wavefunction

No angular dependence

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \quad \frac{1}{r} \frac{\partial^2}{\partial r^2} (ru) = \frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r} = \nabla^2$$

$$\frac{\partial^2}{\partial t^2} \mathbf{E}(r, t) = \frac{1}{\epsilon_0 \mu_0} \frac{1}{r} \frac{\partial^2}{\partial r^2} [r \mathbf{E}(r, t)]$$

4.6 FRAUNHOFER DIFFRACTION

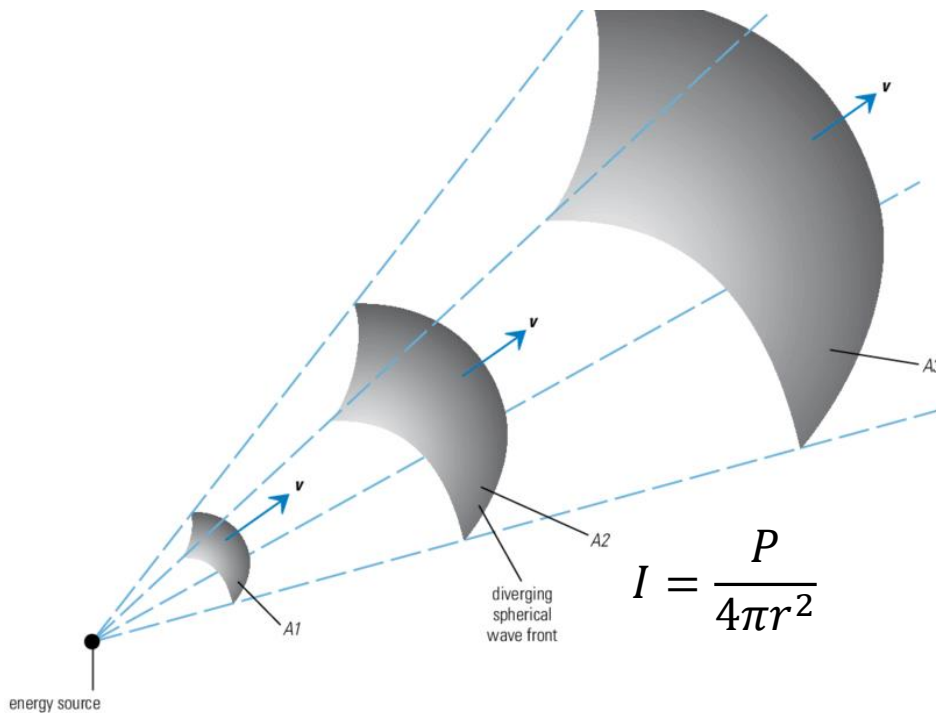
$$\frac{\partial^2}{\partial t^2} \mathbf{E}(r, t) = \frac{1}{\epsilon_0 \mu_0} \frac{1}{r} \frac{\partial^2}{\partial r^2} [r \mathbf{E}(r, t)]$$

$$\frac{\partial^2}{\partial t^2} (rE) = \frac{1}{\epsilon_0 \mu_0} \frac{\partial^2}{\partial r^2} (rE)$$

$$rE(r, t) = E_0 e^{ik \cdot r - i\omega t}$$

$$E(r, t) = \left(\frac{E_0}{r} \right) e^{ik \cdot r - i\omega t} \quad \text{Spherical wavefunction}$$

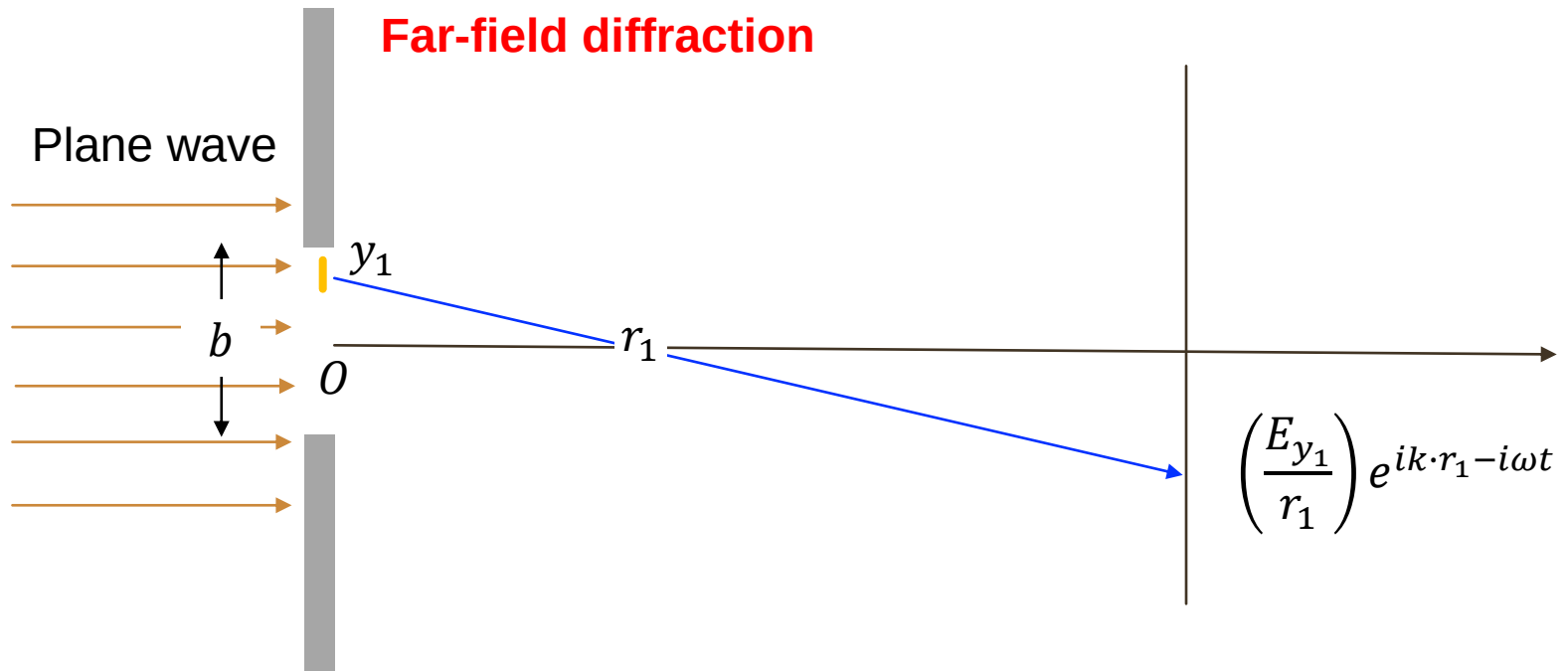
4.6 FRAUNHOFER DIFFRACTION



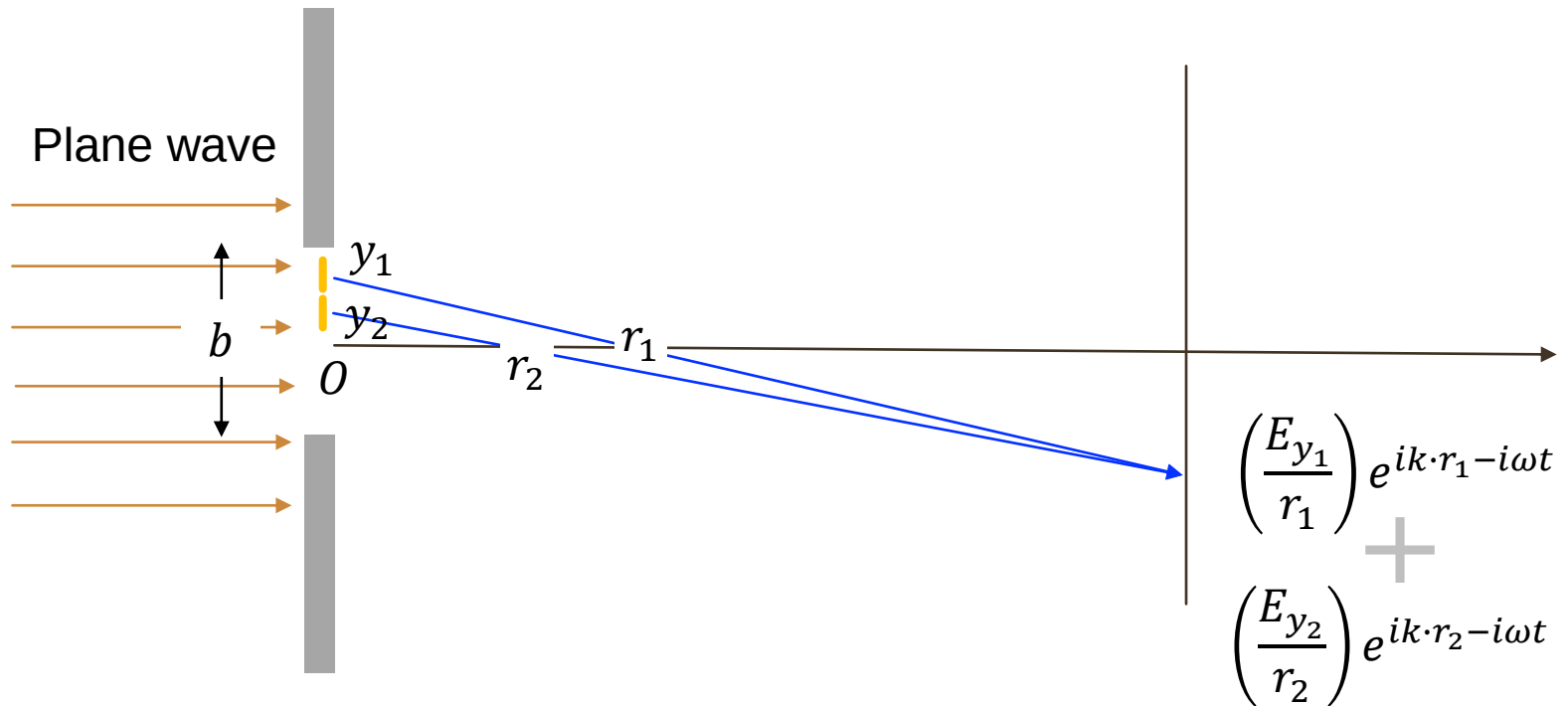
$$\frac{E_{r_1}}{E_{r_2}} = \frac{r_2}{r_1}$$

$$\frac{I_{r_1}}{I_{r_2}} = \frac{r_2^2}{r_1^2}$$

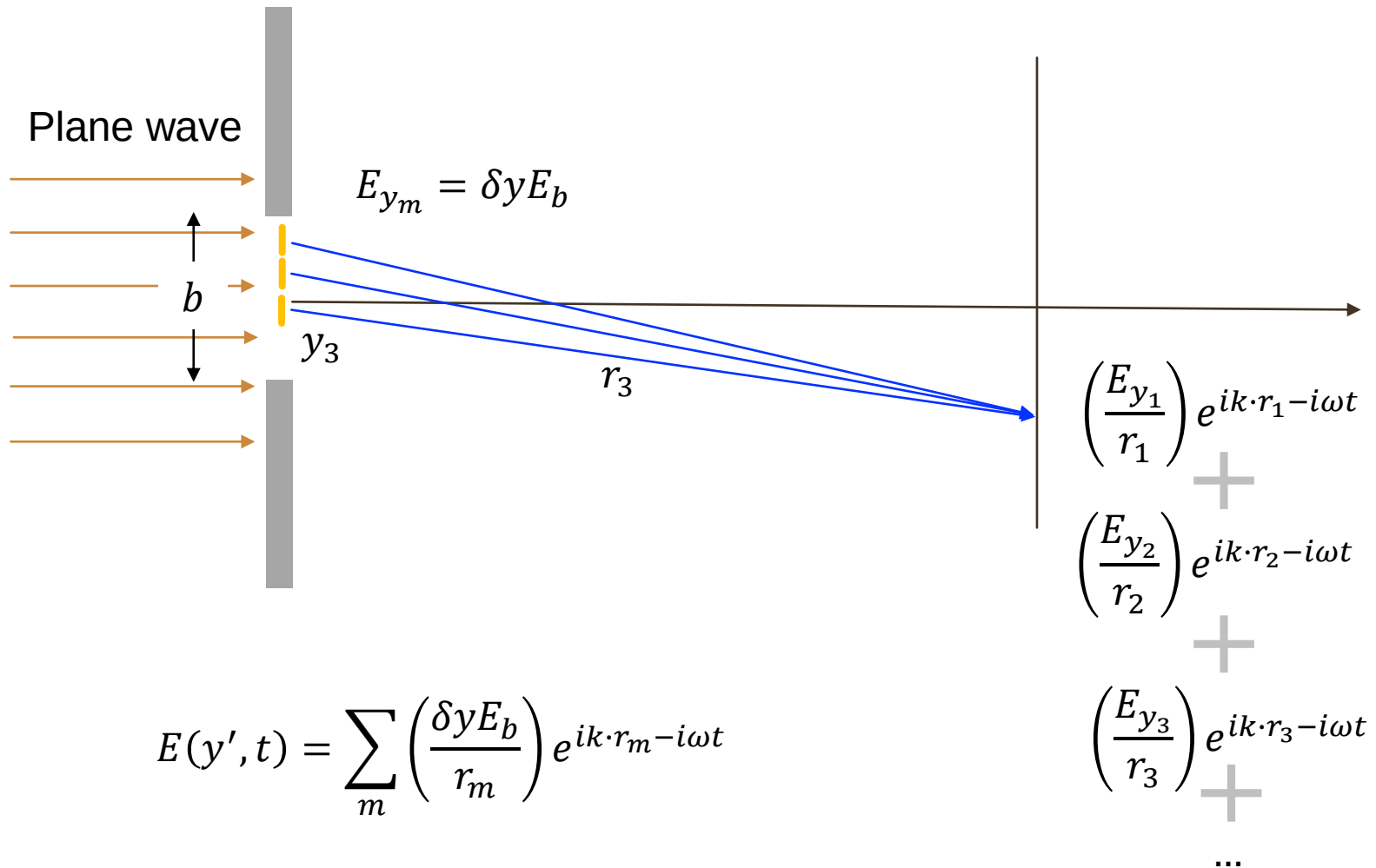
4.6 FRAUNHOFER DIFFRACTION



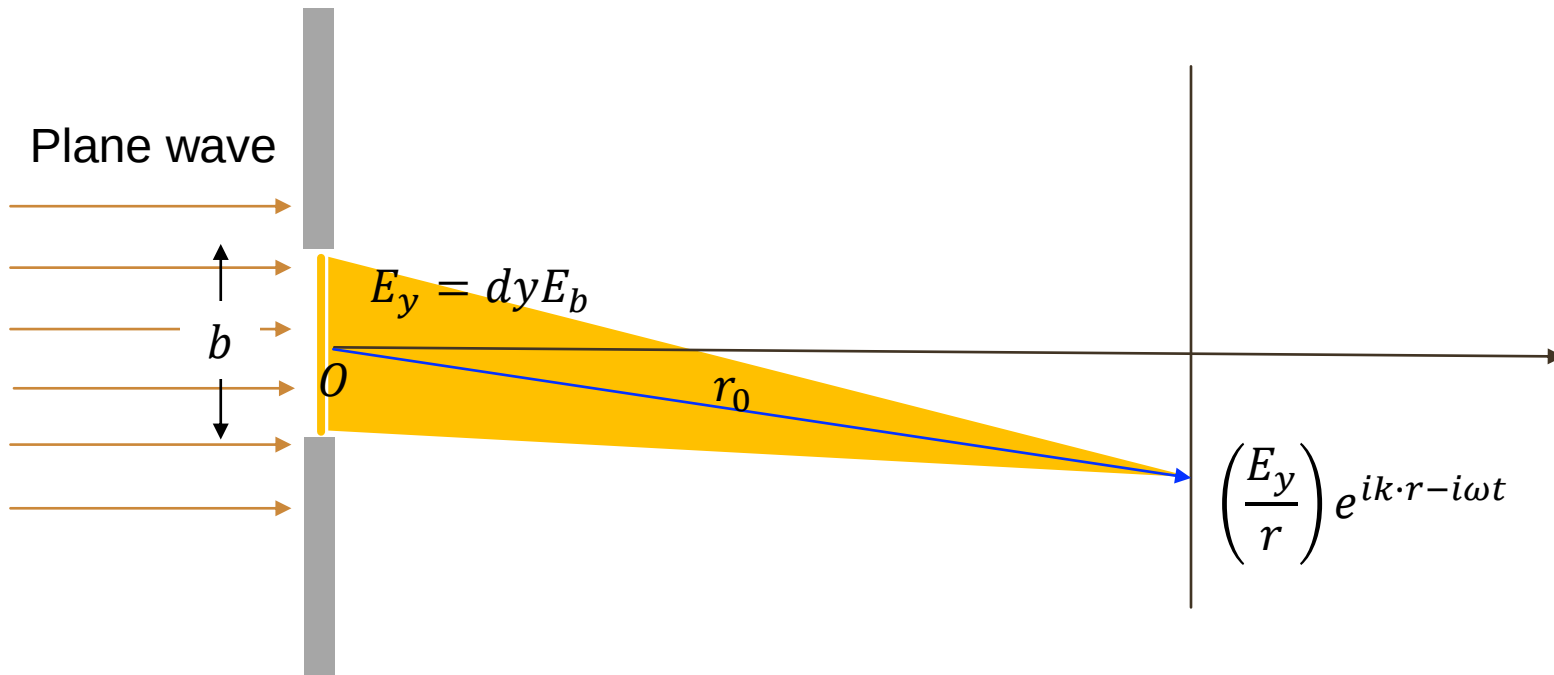
4.6 FRAUNHOFER DIFFRACTION



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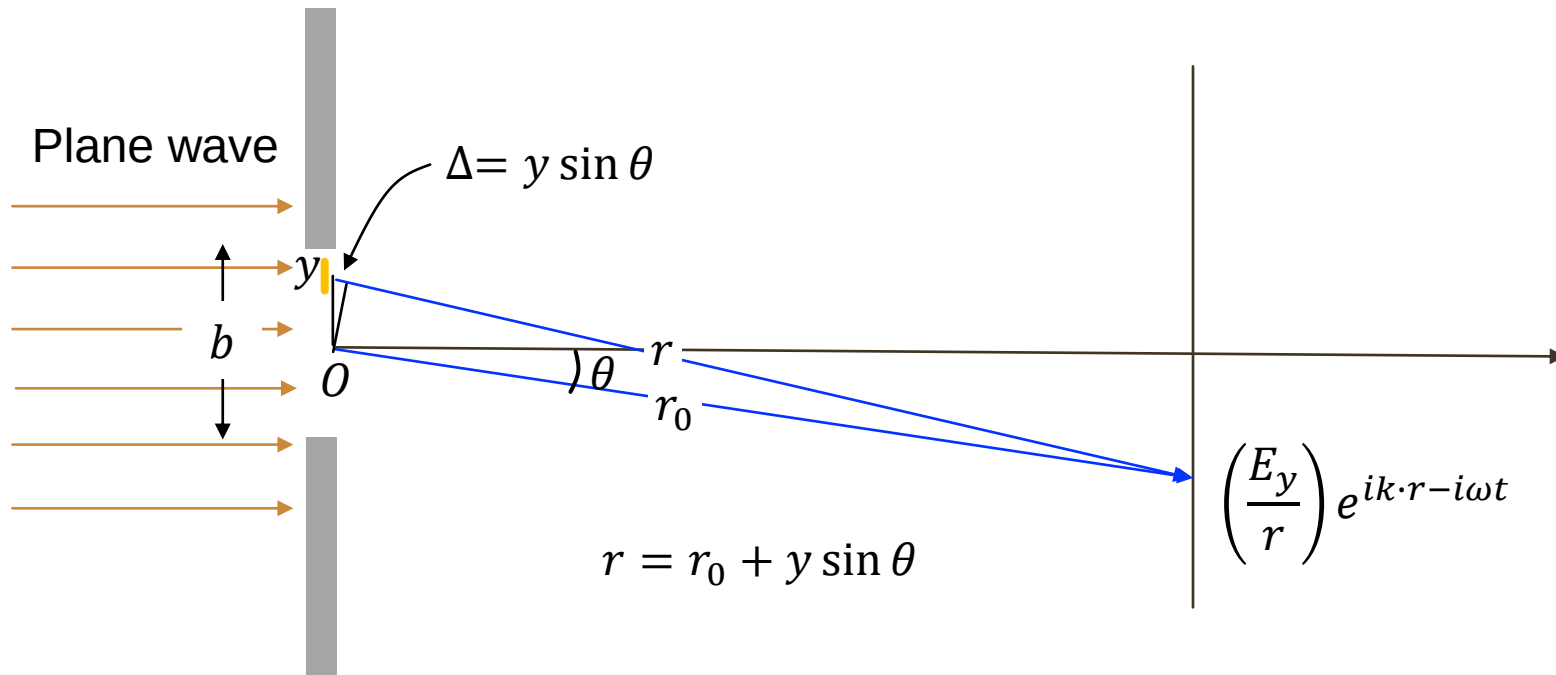


4.6 FRAUNHOFER DIFFRACTION

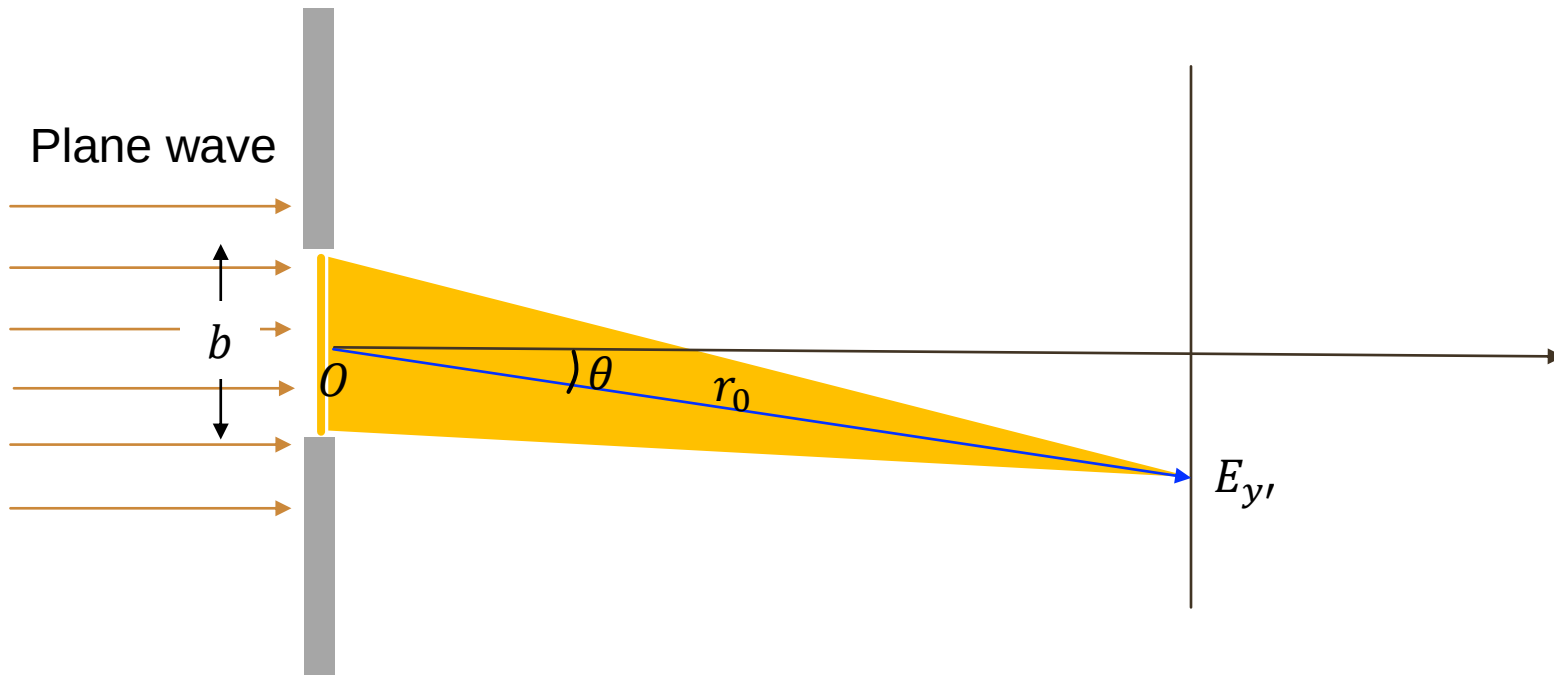


$$E(y', t) = \int_{-\frac{b}{2}}^{\frac{b}{2}} \left(\frac{E_b}{r} \right) e^{ik \cdot \mathbf{r} - i\omega t} dy$$

4.6 FRAUNHOFER DIFFRACTION



4.6 FRAUNHOFER DIFFRACTION



$$E(y', t) = \int_{-\frac{b}{2}}^{\frac{b}{2}} \frac{E_b}{r_0 + y \sin \theta} e^{ikr_0 + iky \sin \theta - i\omega t} dy \approx \frac{E_b e^{ikr_0 - i\omega t}}{r_0} \int_{-\frac{b}{2}}^{\frac{b}{2}} e^{iky \sin \theta} dy$$

4.6 FRAUNHOFER DIFFRACTION

$$E(y', t) = \frac{E_b e^{ikr_0 - i\omega t}}{r_0} \int_{-\frac{b}{2}}^{\frac{b}{2}} e^{iky \sin \theta} dy = \frac{E_b e^{ikr_0 - i\omega t}}{r_0} \frac{e^{i\frac{b}{2}k \sin \theta} - e^{-i\frac{b}{2}k \sin \theta}}{ik \sin \theta}$$

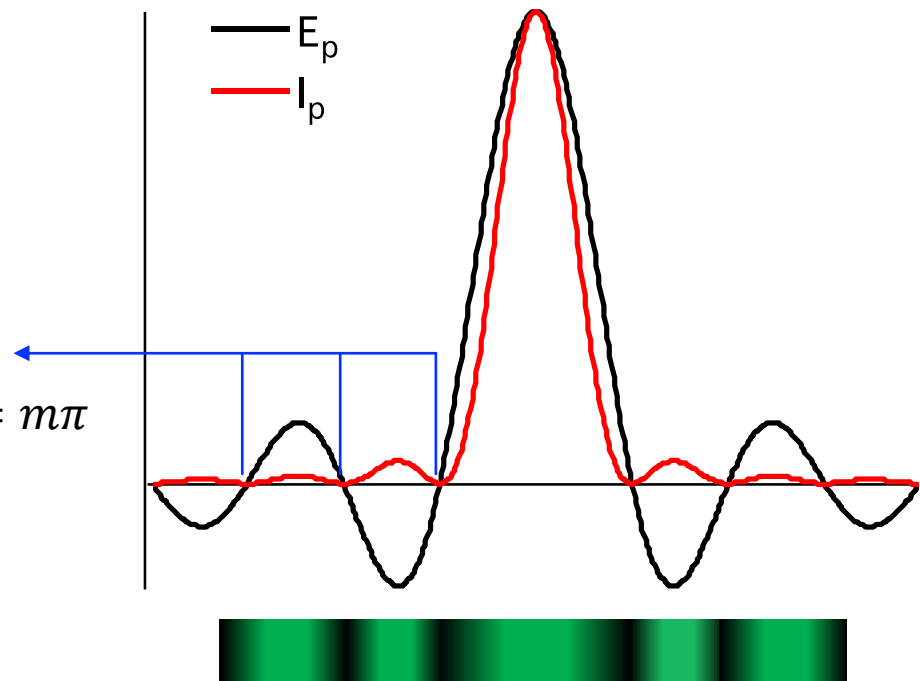
$$\beta = \frac{b}{2} k \sin \theta$$

$$E(y', t) = \frac{b E_b e^{ikr_0 - i\omega t}}{r_0} \frac{\sin \beta}{\beta}$$

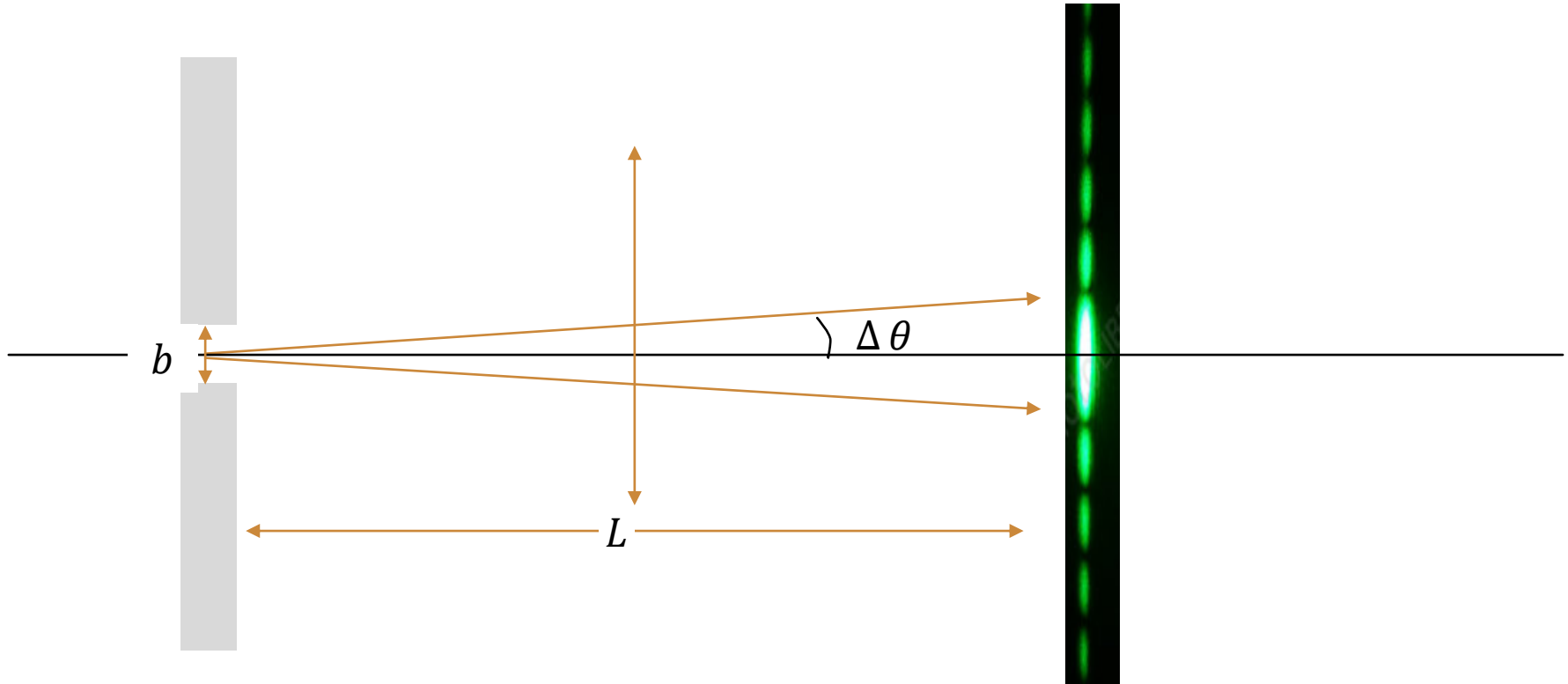
$$I_p = I_0 \frac{\sin^2 \beta}{\beta^2}$$

$$\beta = \frac{b}{2} k \sin \theta = m\pi$$

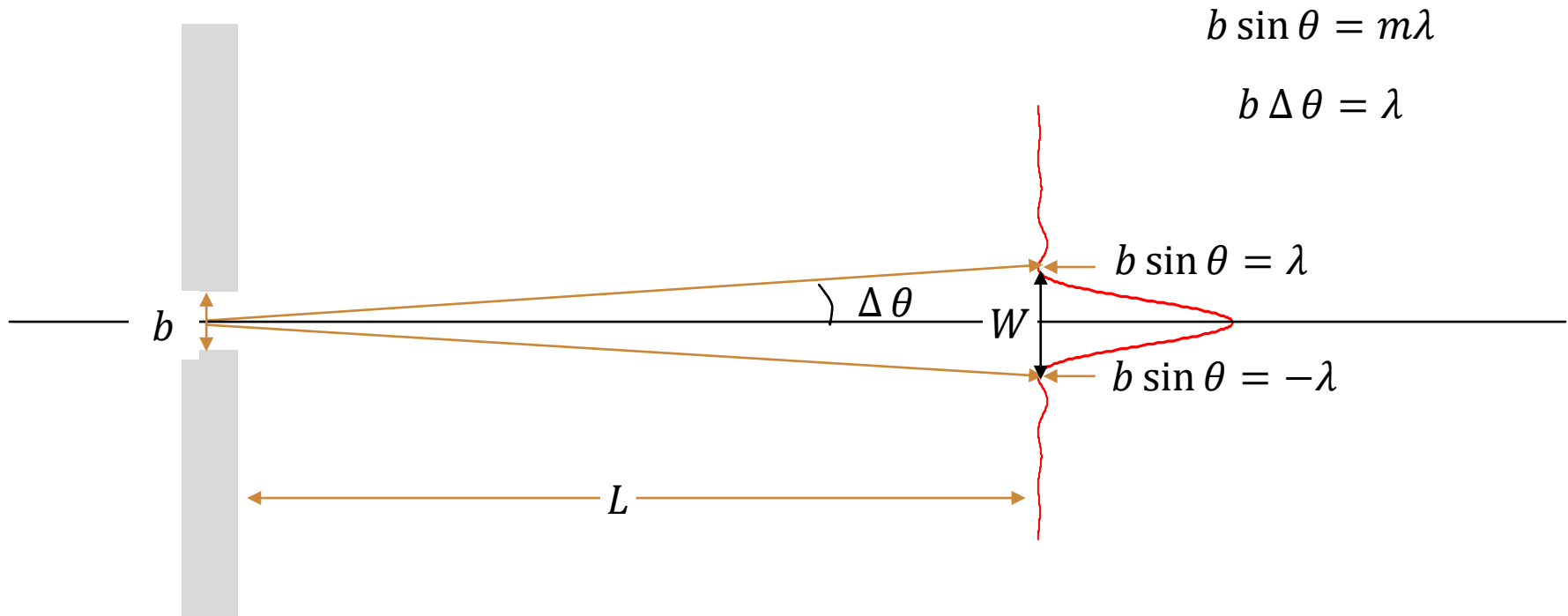
$$b \sin \theta = m\lambda$$



4.6 FRAUNHOFER DIFFRACTION



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$$b \sin \theta = m\lambda$$

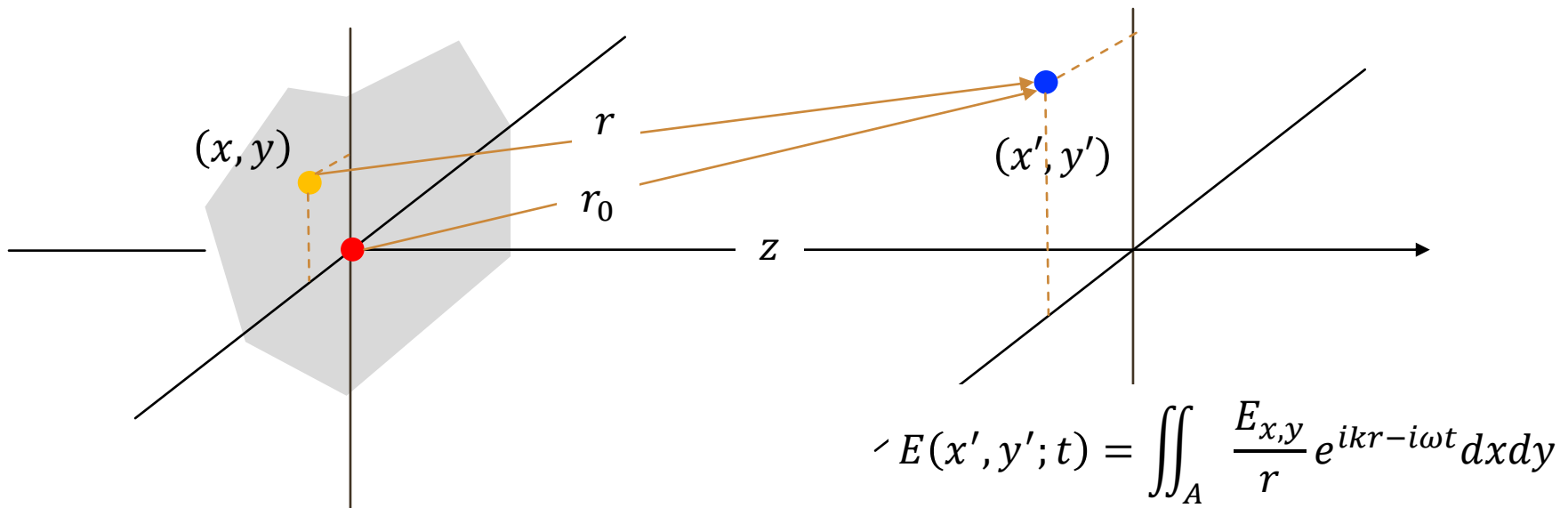
$$b \Delta \theta = \lambda$$

$$b \sin \theta = \lambda$$

$$b \sin \theta = -\lambda$$

Far field beam spreading $W = L \cdot 2\Delta\theta = \frac{2L\lambda}{b}$

4.6 FRAUNHOFER DIFFRACTION



$$r_0 = \sqrt{x'^2 + y'^2 + z^2}$$

$$r = \sqrt{(x - x')^2 + (y - y')^2 + z^2} = \sqrt{r_0^2 - 2xx' - 2yy' + x^2 + y^2}$$

Far field approximation: $r \approx \sqrt{r_0^2 - 2xx' - 2yy'}$

4.6 FRAUNHOFER DIFFRACTION

Taylor expansion at $r = r_0$

$$r(x, y) = r(0,0) + \left. \frac{\partial r}{\partial x} \right|_{x=0,y=0} x + \left. \frac{\partial r}{\partial y} \right|_{x=0,y=0} y + o(x^2, y^2)$$

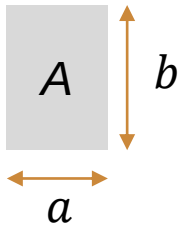
$$r \approx \sqrt{r_0^2 - 2xx' - 2yy'} = r_0 - \frac{2x'}{2r_0}x - \frac{2y'}{2r_0}y + o(x^2, y^2)$$

$$\begin{aligned} E(x', y'; t) &\approx \iint_A \frac{E_{x,y}}{r} e^{ikr - i\omega t} dx dy = \iint_A \frac{E_{x,y}}{r_0 - \frac{x'}{r_0}x - \frac{y'}{r_0}y} e^{ik\left(r_0 - \frac{x'}{r_0}x - \frac{y'}{r_0}y\right) - i\omega t} dx dy \\ &\approx \iint_A \frac{E_{x,y}}{r_0} e^{ikr_0 - i\omega t} e^{-ik\left(\frac{x'}{r_0}x + \frac{y'}{r_0}y\right)} dx dy \end{aligned}$$

$$r_0 \approx z$$

$$E(x', y'; t) \approx \iint_A \frac{E_{x,y}}{z} e^{ikr_0 - i\omega t} e^{-ik\left(\frac{x'}{z}x + \frac{y'}{z}y\right)} dx dy$$

4.6 FRAUNHOFER DIFFRACTION



$$E(x', y'; t) = \frac{E_{x,y}}{z} e^{ikz - i\omega t} \int_{-\frac{a}{2}}^{\frac{a}{2}} \int_{-\frac{b}{2}}^{\frac{b}{2}} e^{-ik\frac{x'}{z}x - ik\frac{y'}{z}y} dx dy$$

$$= \frac{E_{x,y}}{z} e^{ikz - i\omega t} \frac{e^{i\frac{a}{2}k\frac{x'}{z}} - e^{-i\frac{a}{2}k\frac{x'}{z}}}{ik\frac{x'}{z}} \frac{e^{i\frac{b}{2}k\frac{y'}{z}} - e^{-i\frac{b}{2}k\frac{y'}{z}}}{ik\frac{y'}{z}}$$

$$\alpha = \frac{a}{2} k \frac{x'}{r_0}$$

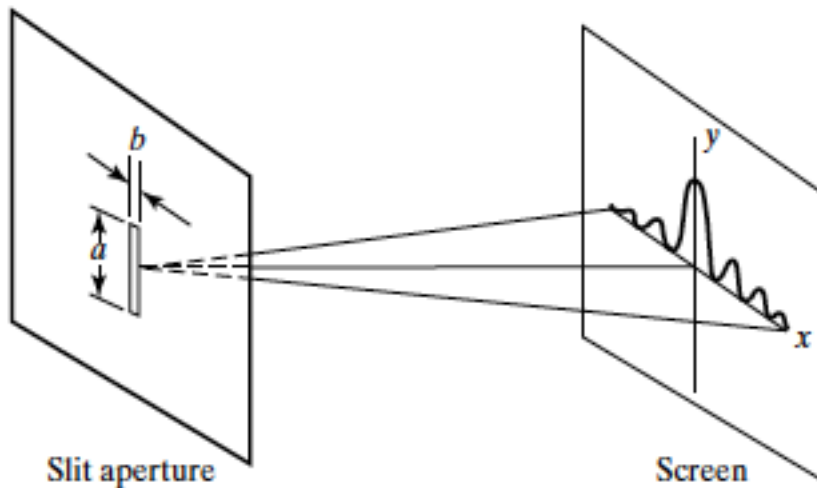
$$\beta = \frac{b}{2} k \frac{y'}{r_0}$$

$$E(x', y'; t) = \frac{abE_{x,y}}{z} e^{ikz - i\omega t} \frac{\sin \alpha}{\alpha} \frac{\sin \beta}{\beta}$$

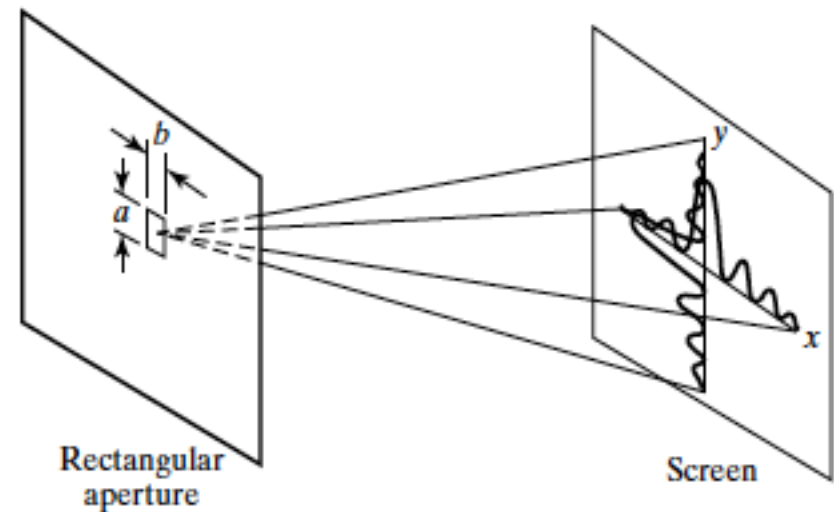
$$I_p = I_0 \frac{\sin^2 \alpha}{\alpha^2} \frac{\sin^2 \beta}{\beta^2}$$

4.6 FRAUNHOFER DIFFRACTION

Rectangular aperture



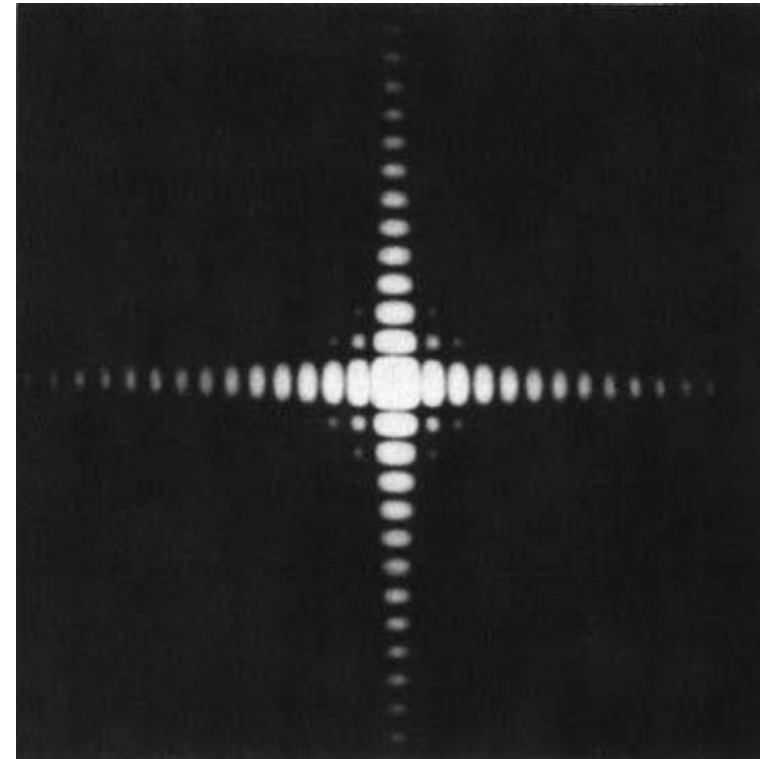
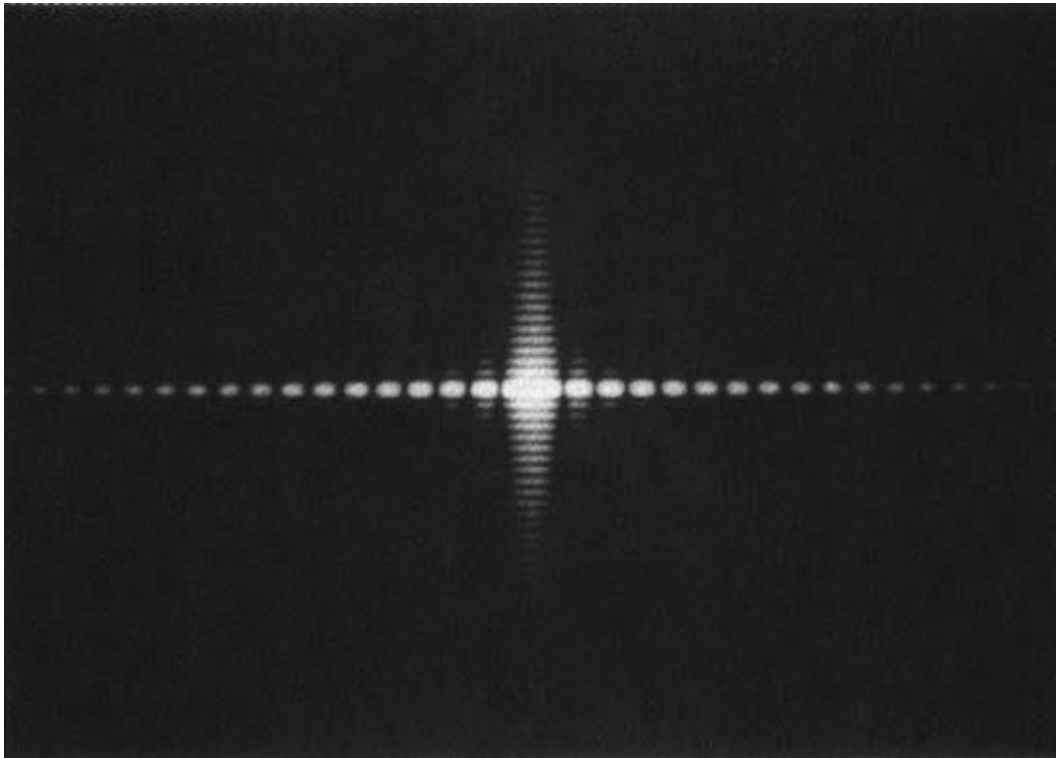
$$I_p = I_0 \frac{\sin^2 \beta}{\beta^2}$$



$$I_p = I_0 \frac{\sin^2 \alpha}{\alpha^2} \frac{\sin^2 \beta}{\beta^2}$$

4.6 FRAUNHOFER DIFFRACTION

Rectangular aperture

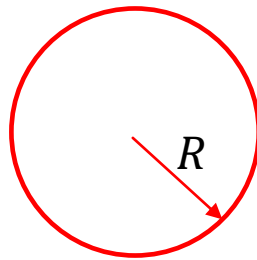


4.6 FRAUNHOFER DIFFRACTION

Circular aperture

$$I_p = I_0 \left(\frac{2J_1(\gamma)}{\gamma} \right)^2$$

$$\gamma = kR \sin \theta$$

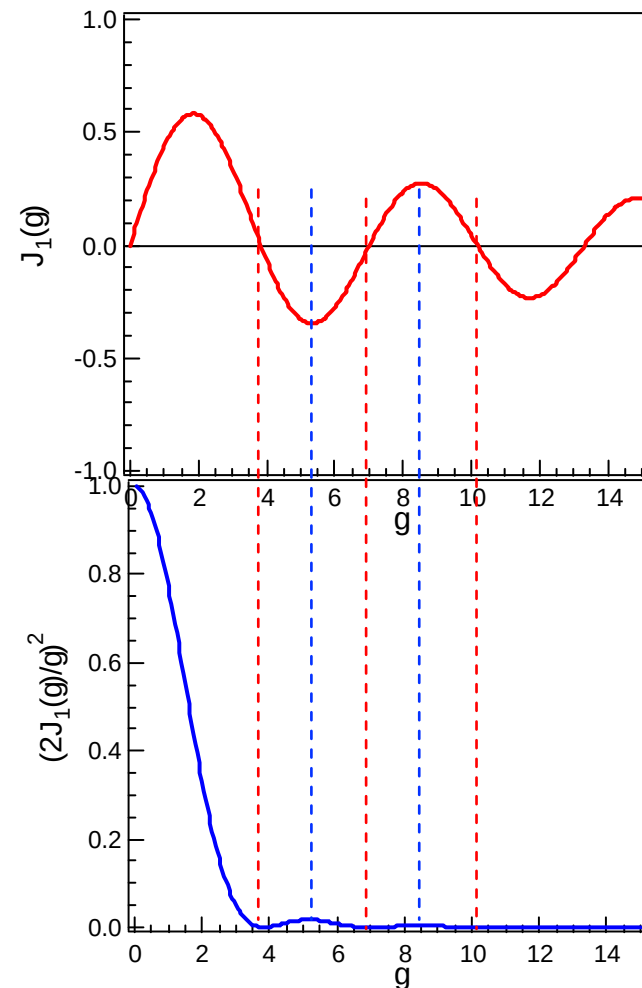


First order Bessel function of the first kind

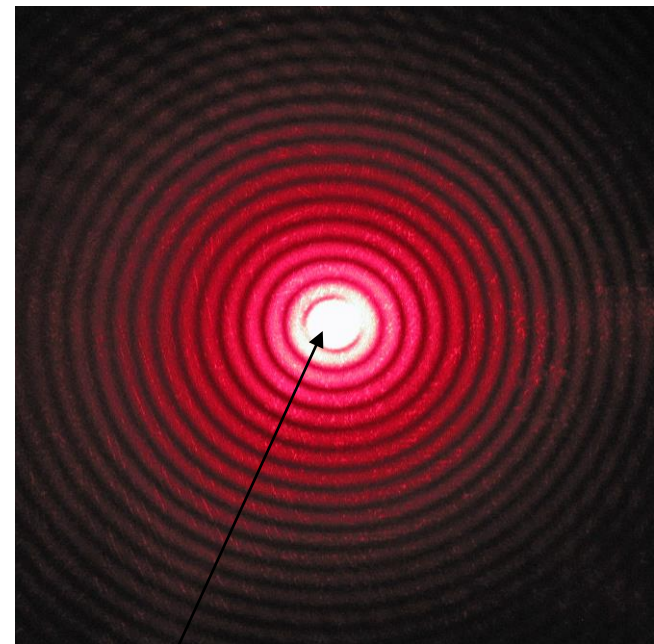
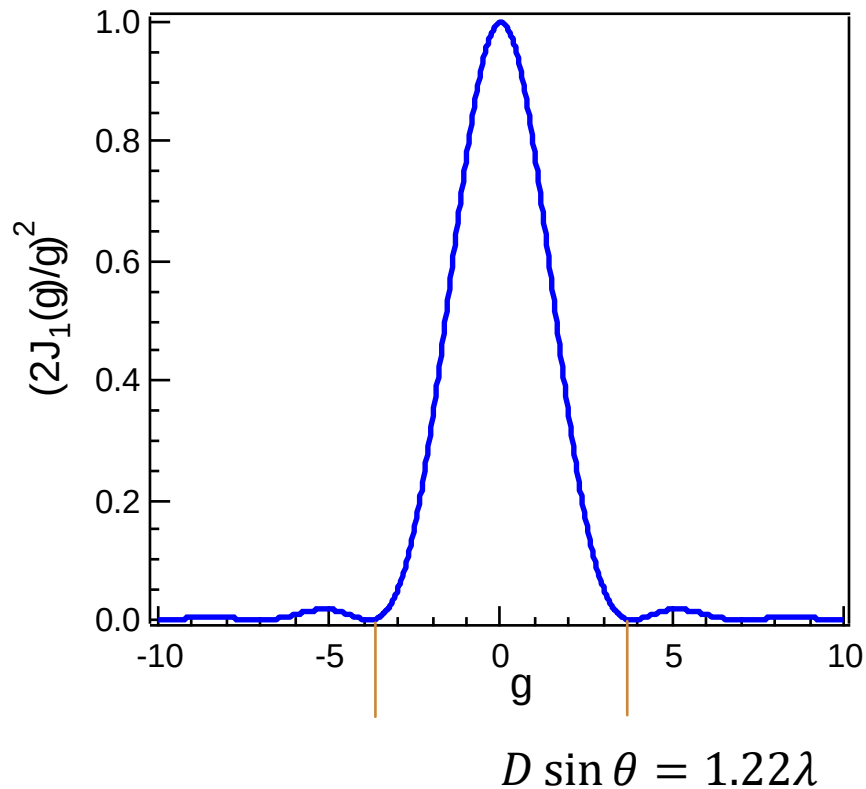
$$J_1(\gamma) = \frac{\gamma}{2} - \frac{\left(\frac{\gamma}{2}\right)^3}{1^2 \cdot 2} + \frac{\left(\frac{\gamma}{2}\right)^5}{1^2 \cdot 2^2 \cdot 3} - \dots$$

$$J_1 = 0 \quad \gamma = kR \sin \theta = \frac{\pi D}{\lambda} \sin \theta = 3.83$$

$$D \sin \theta = 1.22\lambda$$



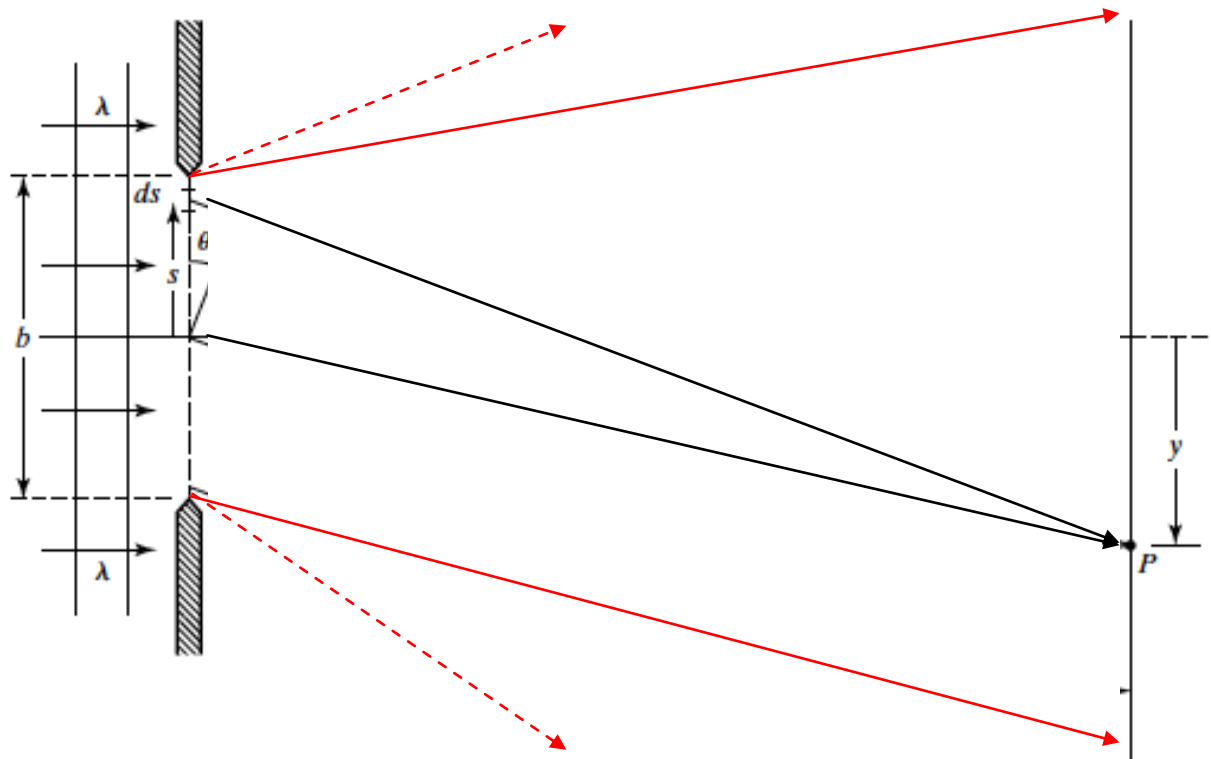
4.6 FRAUNHOFER DIFFRACTION



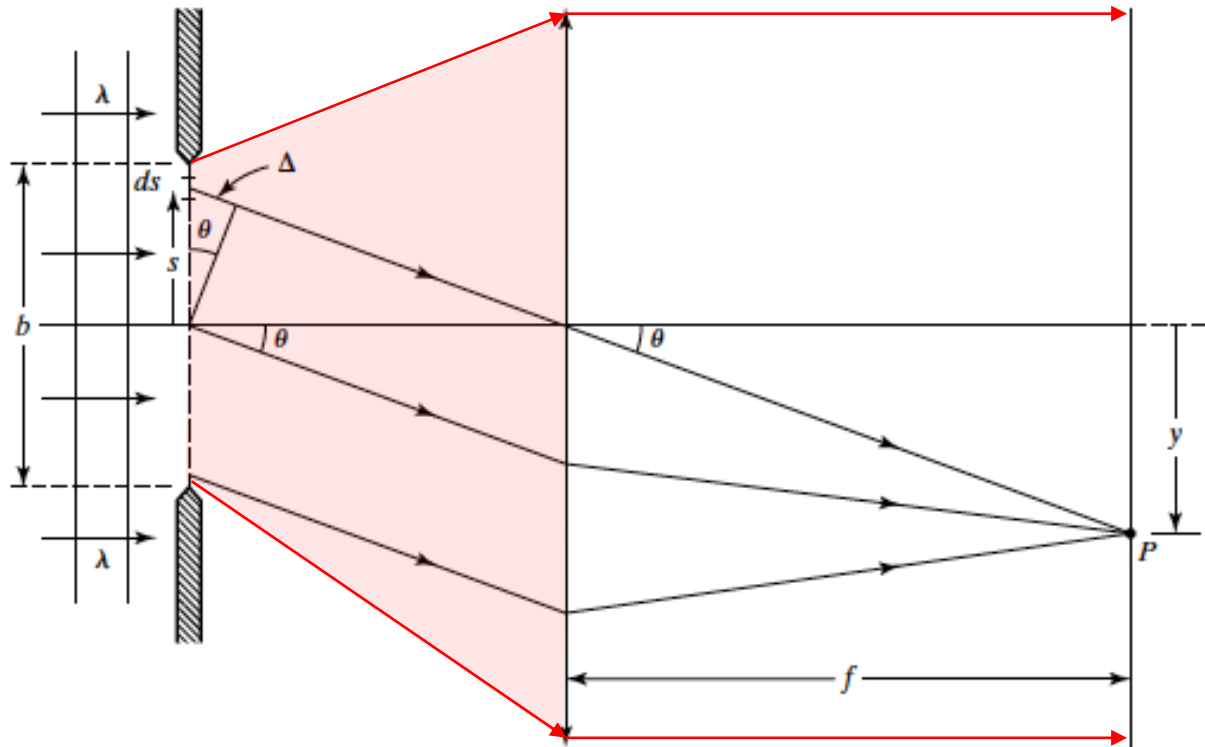
<https://en.wikipedia.org/wiki/Diffraction>

Airy disc

4.6 FRAUNHOFER DIFFRACTION

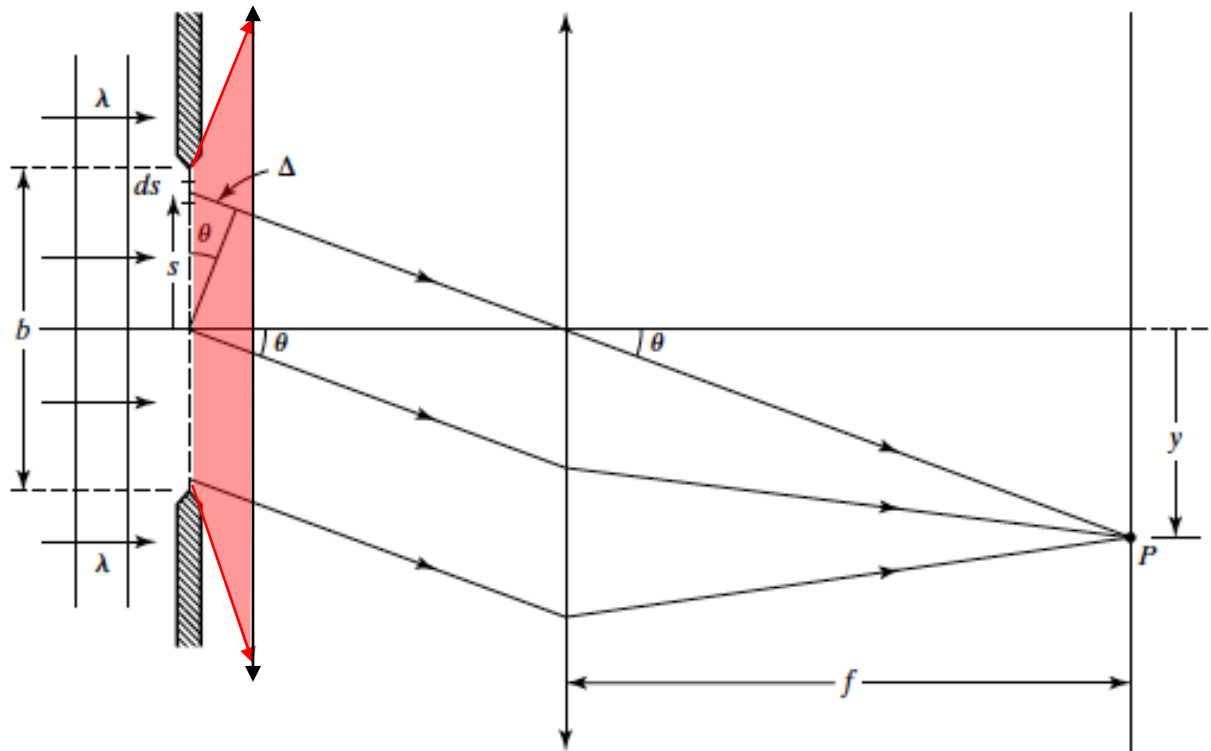


4.6 FRAUNHOFER DIFFRACTION



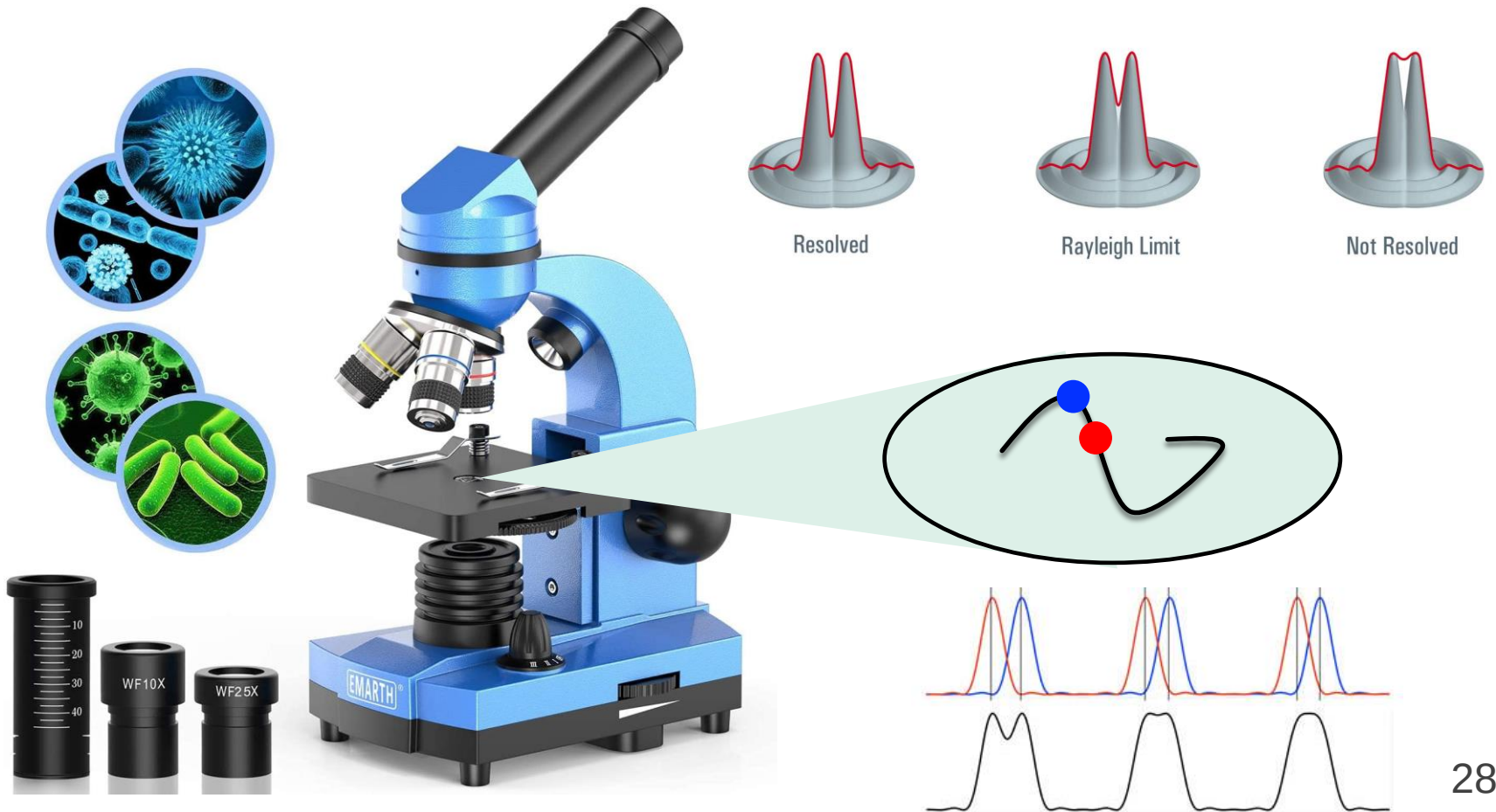
More light is collected, but physics does not change

4.6 FRAUNHOFER DIFFRACTION

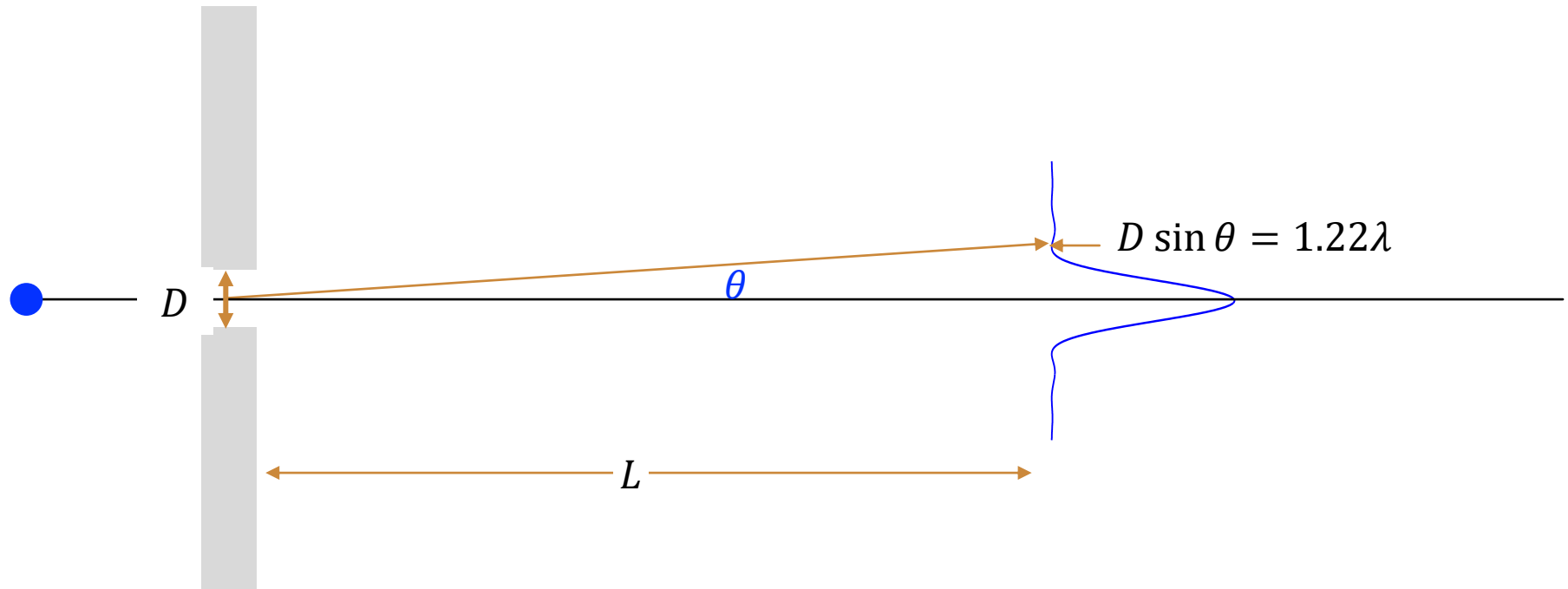


4.6 FRAUNHOFER DIFFRACTION

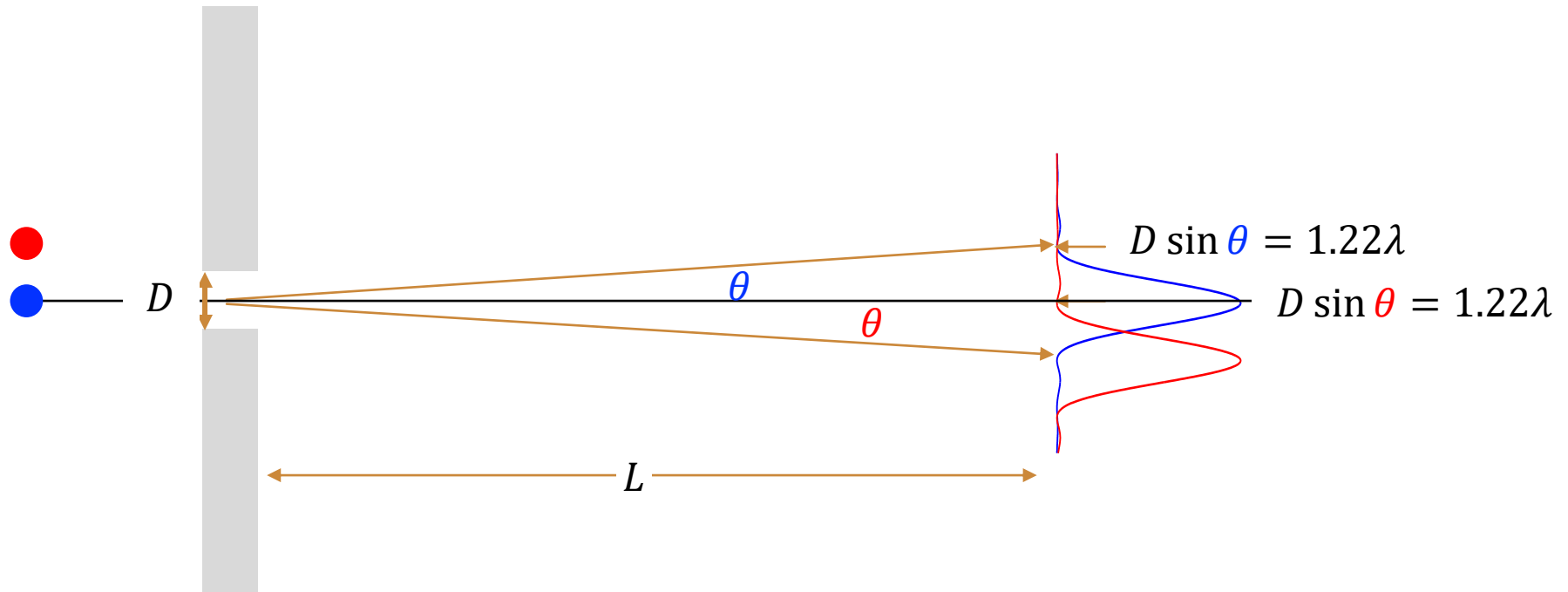
Resolution



4.6 FRAUNHOFER DIFFRACTION

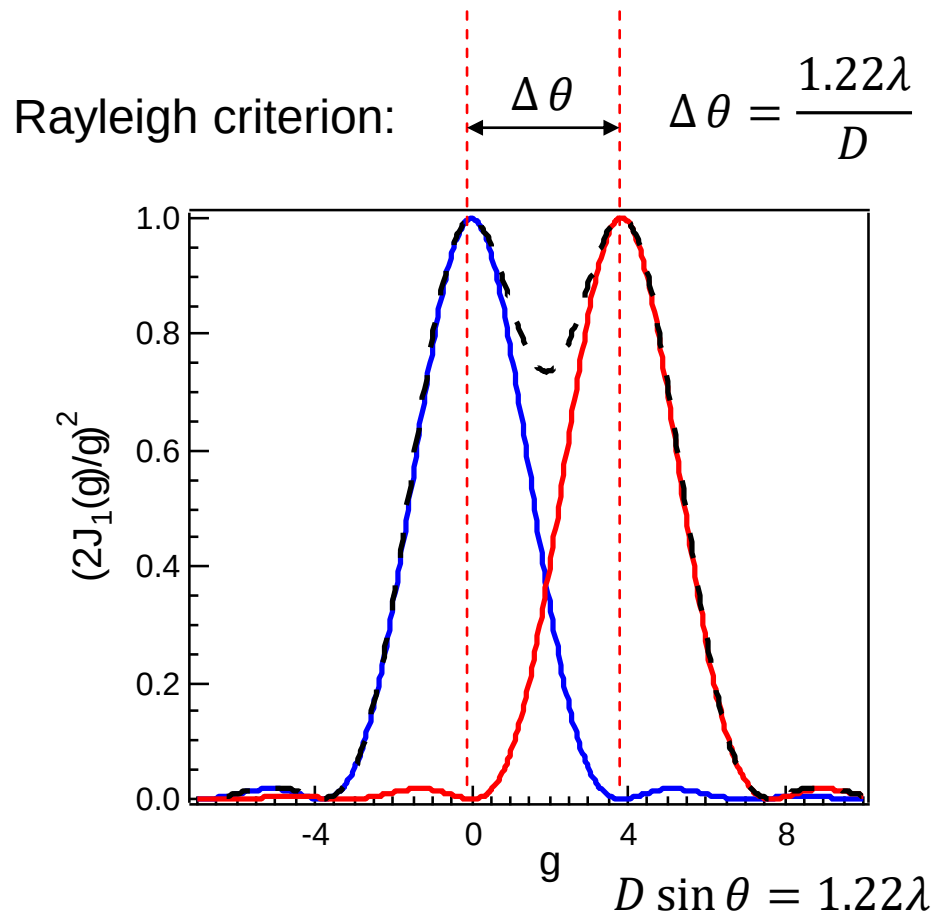


4.6 FRAUNHOFER DIFFRACTION



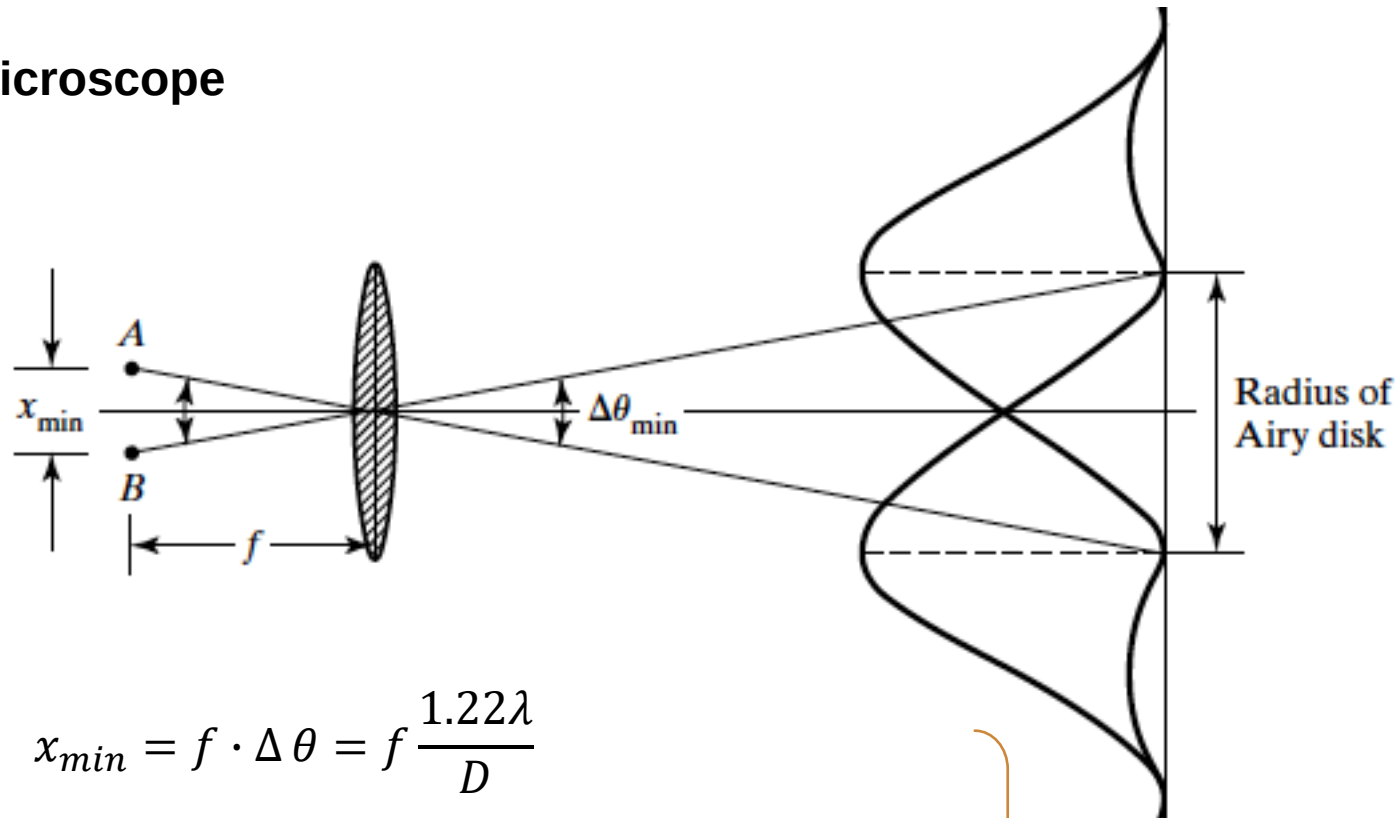
4.6 FRAUNHOFER DIFFRACTION

Angular resolution



4.6 FRAUNHOFER DIFFRACTION

Microscope



$$x_{\min} = f \cdot \Delta\theta = f \frac{1.22\lambda}{D}$$

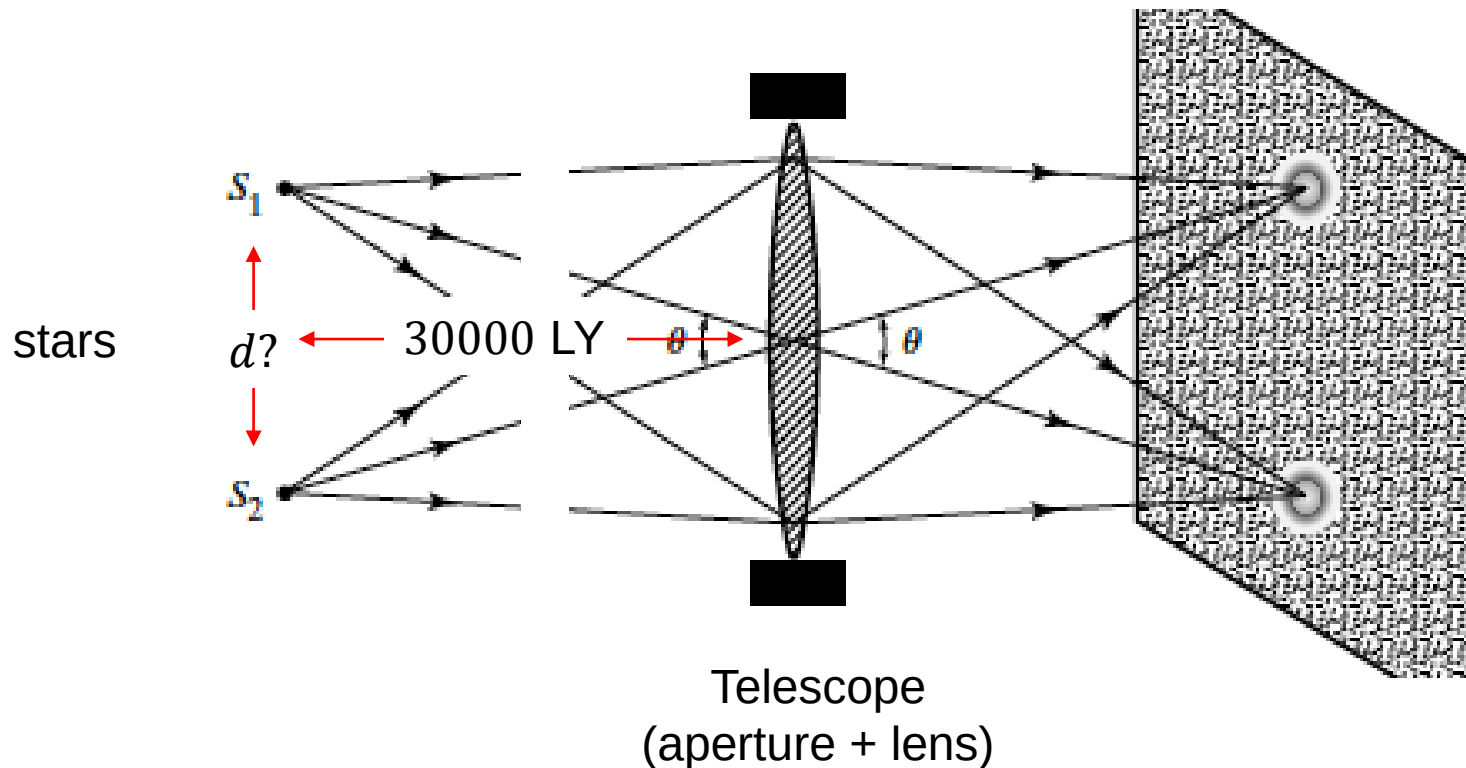
$$\frac{D}{f} \sim 2NA \text{ (numerical aperture)}$$

$$x_{\min} \sim \lambda$$

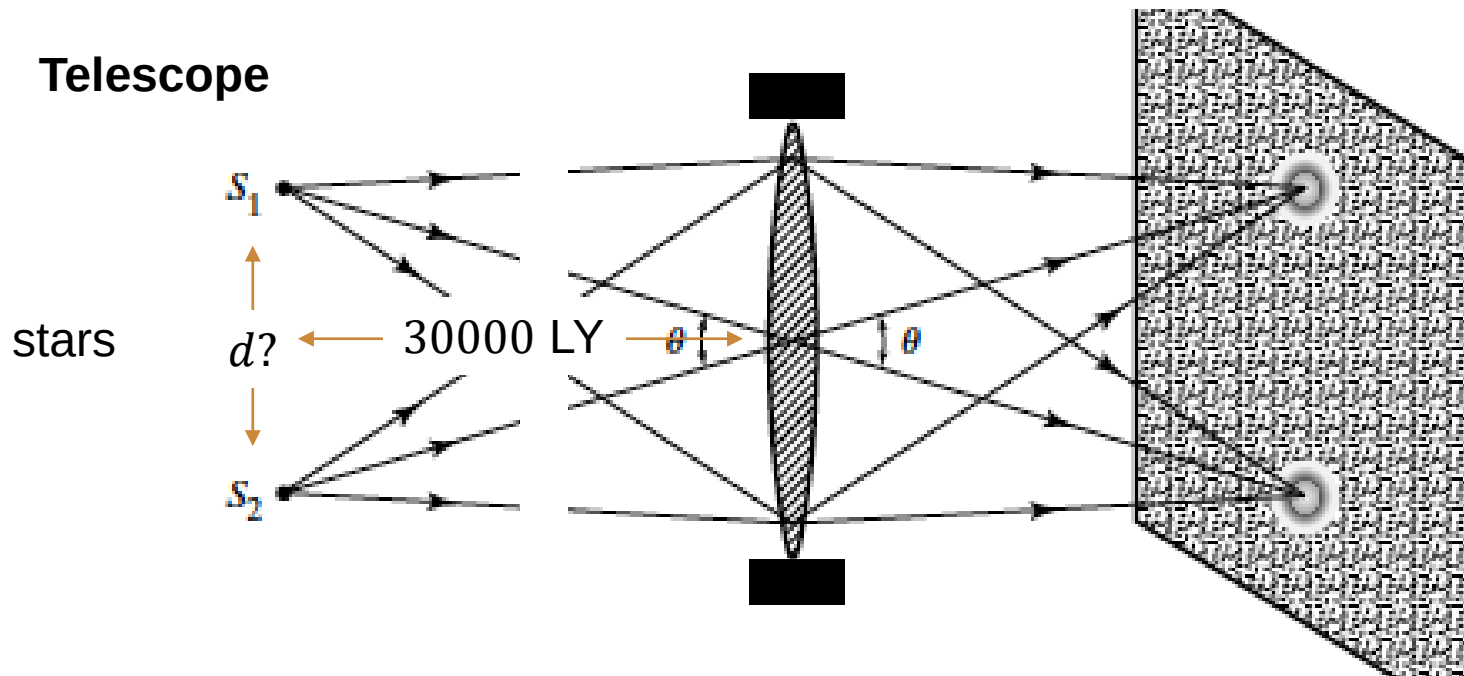
NA is ~ 1.2 - 1.5 for a good oil-immersion objective

4.6 FRAUNHOFER DIFFRACTION

Diffraction limit



4.6 FRAUNHOFER DIFFRACTION



$$\Delta \theta = \frac{1.22\lambda}{D}$$

$$\lambda = 550\text{nm}$$

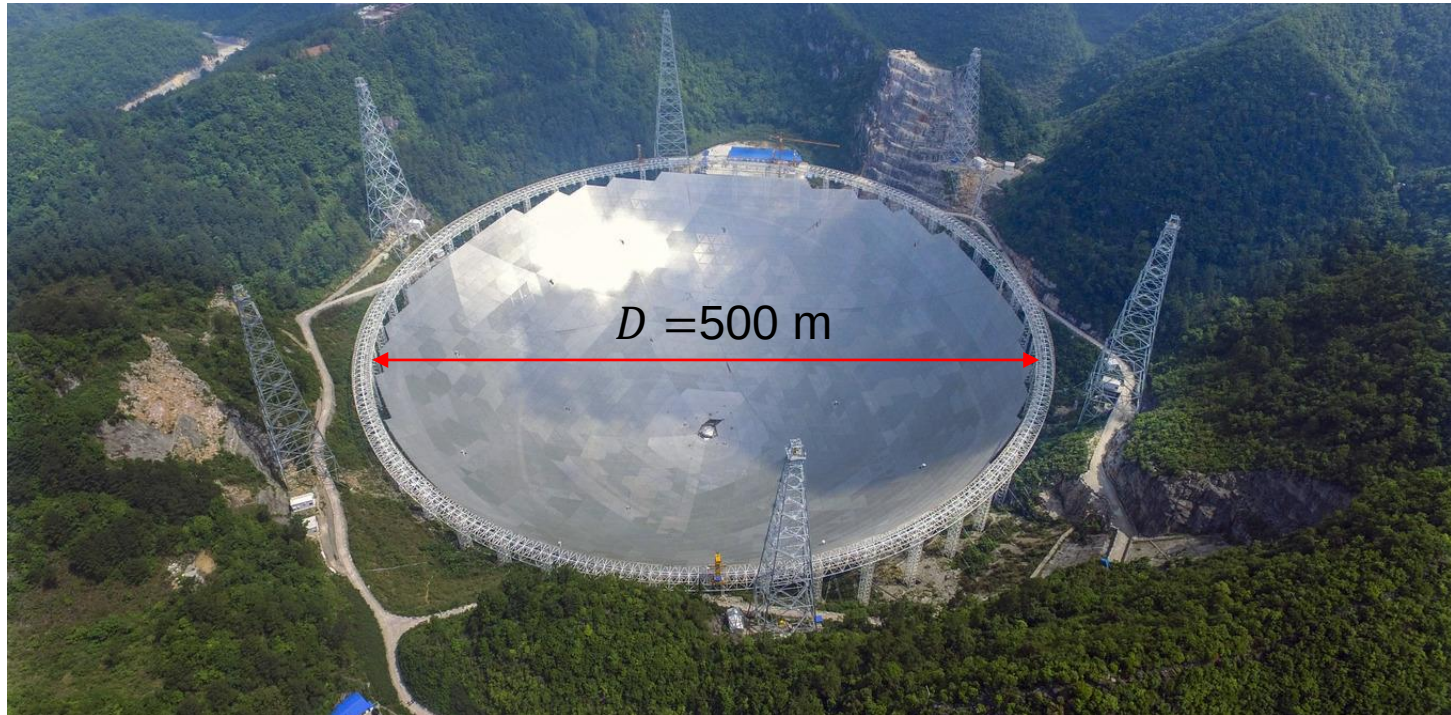
$$D = 35\text{mm}$$

$$\Delta \theta = 1.92 \times 10^{-5} \text{ rad}$$

$$d = 30000\text{LY} * \Delta \theta = 0.58\text{LY}$$

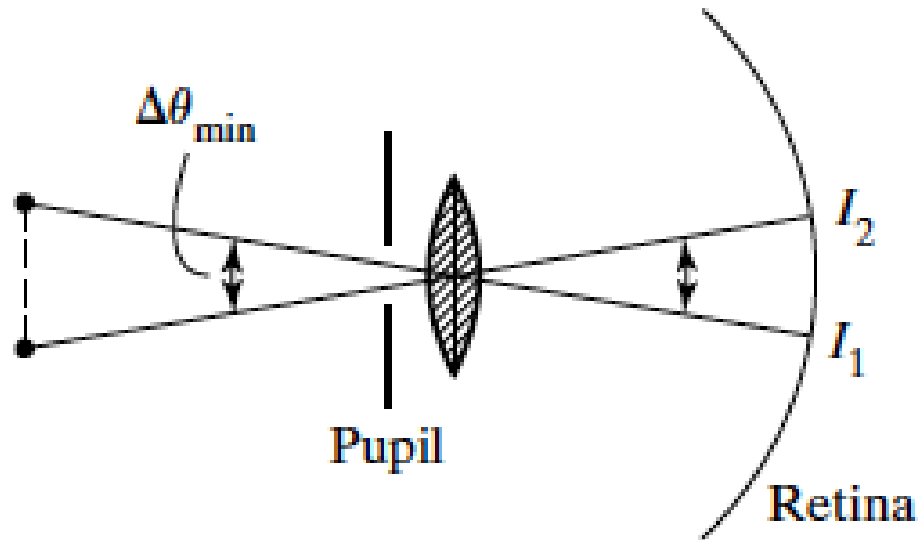
$$\Delta \theta \downarrow \quad D \uparrow \quad \lambda \downarrow$$

4.6 FRAUNHOFER DIFFRACTION



4.6 FRAUNHOFER DIFFRACTION

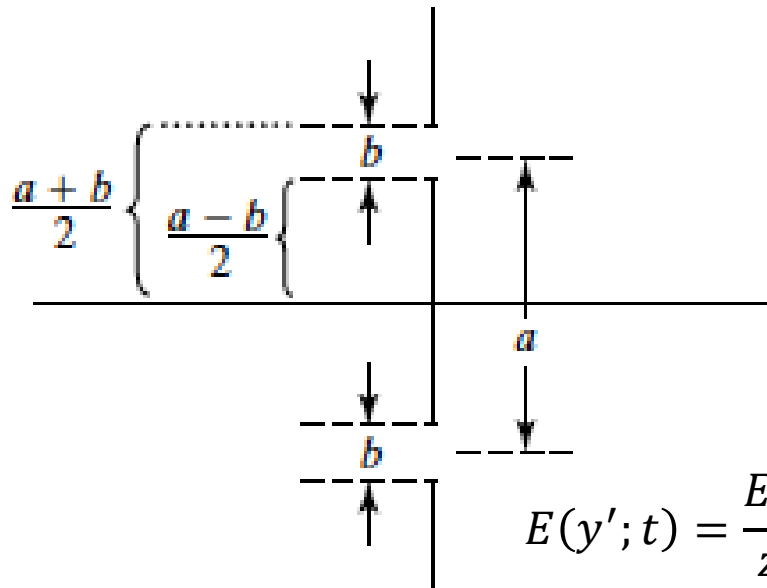
Eye



$$x_{\min} = f \frac{1.22\lambda}{D}$$

4.6 FRAUNHOFER DIFFRACTION

Double-slit diffraction

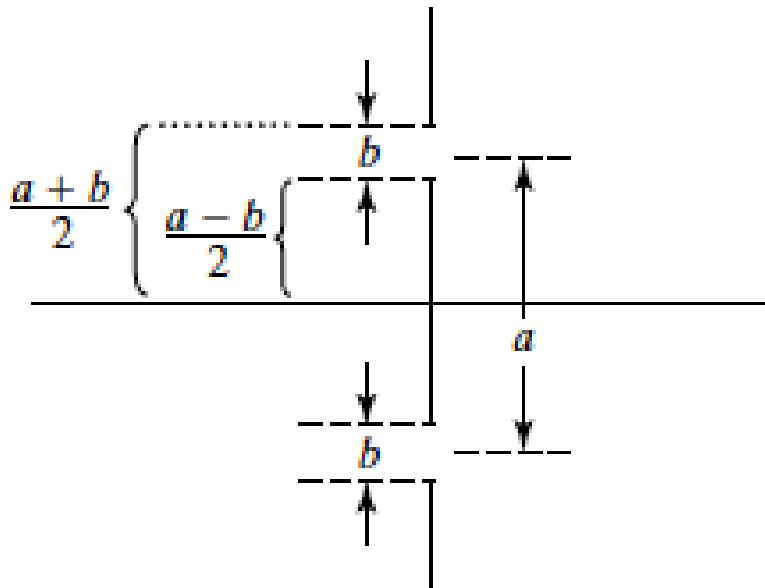


$$E(y'; t) = \frac{E_y}{z} e^{ikz - i\omega t} \left[\int_{-\frac{a+b}{2}}^{-\frac{a-b}{2}} e^{iky \sin \theta} dy + \int_{\frac{a-b}{2}}^{\frac{a+b}{2}} e^{iky \sin \theta} dy \right]$$

$$= \frac{E_y}{z} e^{ikz - i\omega t} \frac{1}{ik \sin \theta} \left[e^{-i\frac{a-b}{2}k \sin \theta} - e^{-i\frac{a+b}{2}k \sin \theta} + e^{i\frac{a+b}{2}k \sin \theta} - e^{i\frac{a-b}{2}k \sin \theta} \right]$$

4.6 FRAUNHOFER DIFFRACTION

Double-slit diffraction

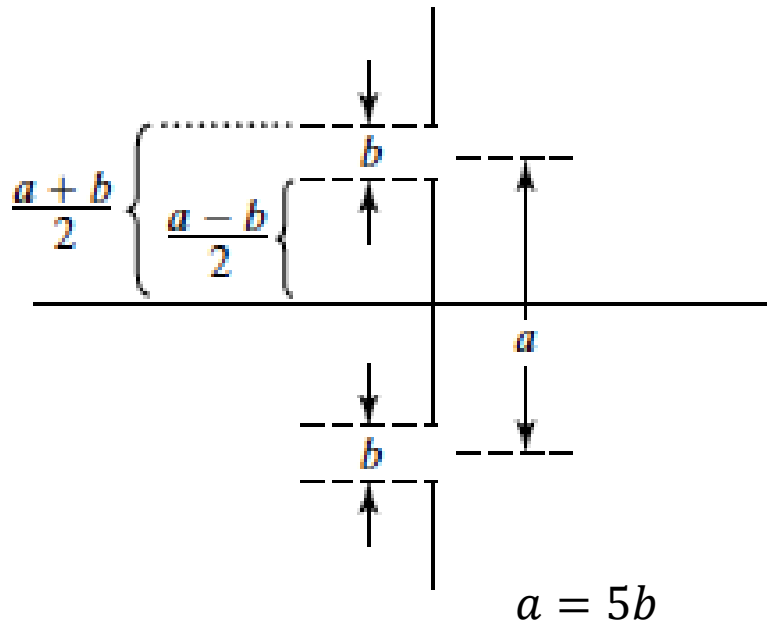


$$\alpha = \frac{ka \sin \theta}{2} \quad \beta = \frac{kb \sin \theta}{2}$$

$$E(y'; t) = \frac{E_y}{z} e^{ikz - i\omega t} \frac{b}{2i\beta} [e^{-i\alpha}(e^{i\beta} - e^{-i\beta}) + e^{i\alpha}(e^{i\beta} - e^{-i\beta})] = \frac{2bE_y}{z} e^{ikz - i\omega t} \cos \alpha \frac{\sin \beta}{\beta}$$

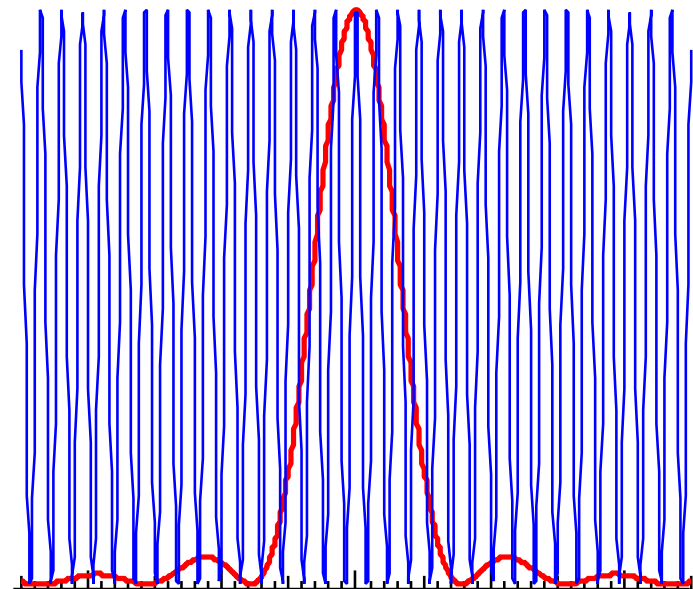
4.6 FRAUNHOFER DIFFRACTION

Double-slit diffraction



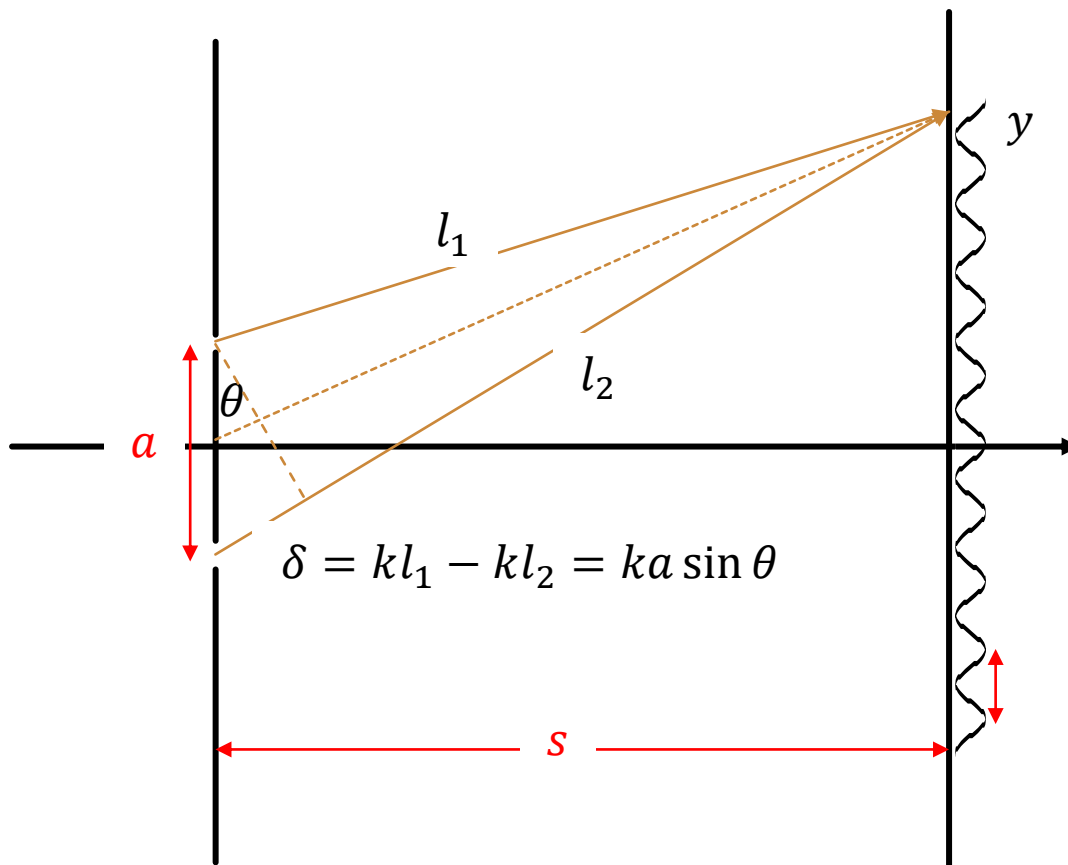
$$E(y'; t) = \frac{2bE_y}{z} e^{ikz - i\omega t} \cos \alpha \frac{\sin \beta}{\beta}$$

$$I(y') = 4I_0 \cos^2 \alpha \frac{\sin^2 \beta}{\beta^2}$$



REVIEW

Young's double-slit experiment



$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$$

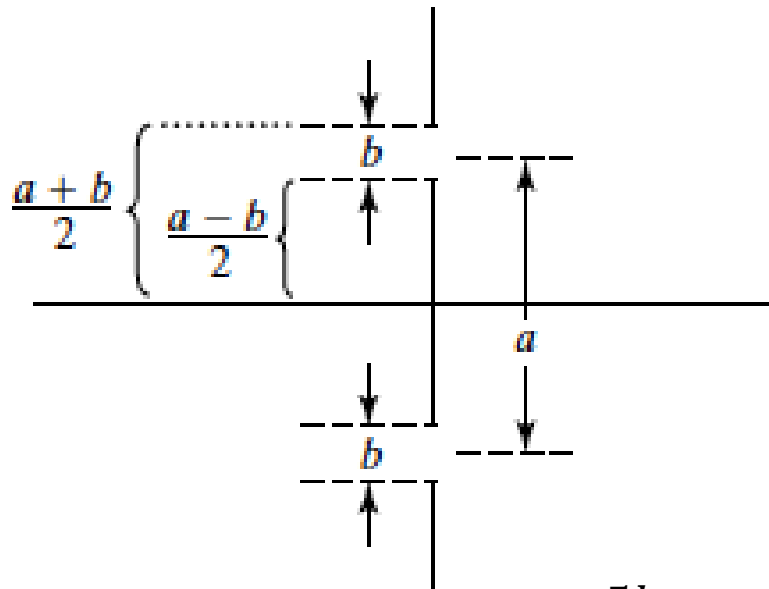
$$I = 2I_0 + 2I_0 \cos \delta$$

$$I = 2I_0(1 + \cos \delta) = 4I_0 \cos^2 \frac{\delta}{2}$$

$$\frac{\delta}{2} = k \frac{a}{2} \sin \theta$$

4.6 FRAUNHOFER DIFFRACTION

Double-slit diffraction



$$a = 5b$$

diffraction minima:

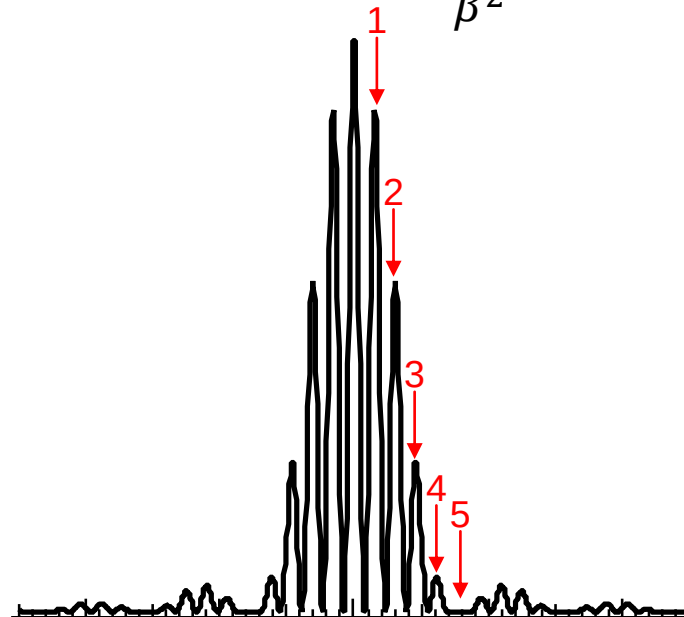
$$\frac{kb \sin \theta}{2} = m\pi$$

Interference fringe maxima

$$\frac{ka \sin \theta}{2} = p\pi$$

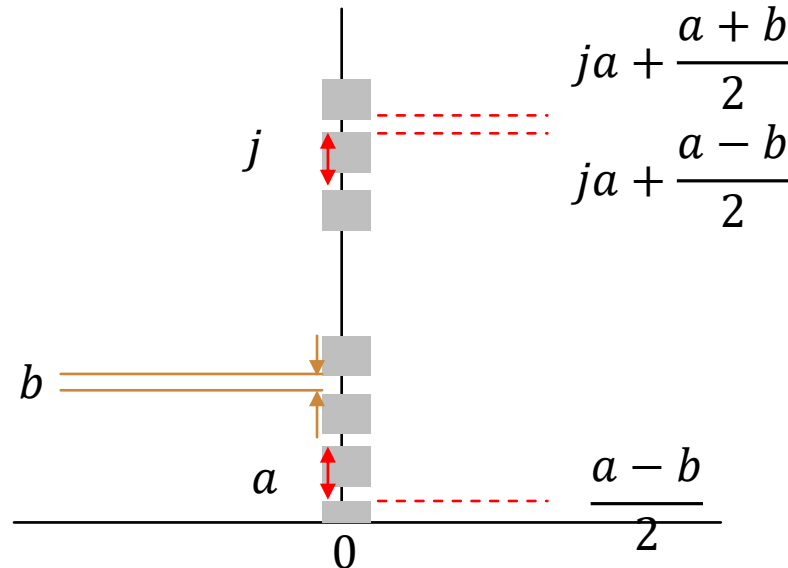
$$E(y'; t) = \frac{2bE_y}{z} e^{ikz - i\omega t} \cos \alpha \frac{\sin \beta}{\beta}$$

$$I(y') = 4I_0 \cos^2 \alpha \frac{\sin^2 \beta}{\beta^2}$$



4.6 FRAUNHOFER DIFFRACTION

Many-slit diffraction



$$E(y'; t) = \frac{E_y}{z} e^{ikz - i\omega t} \sum_{j=0}^{j=N/2} \left[\int_{-\frac{(2j+1)a+b}{2}}^{-\frac{(2j+1)a-b}{2}} e^{iky \sin \theta} dy + \int_{\frac{(2j+1)a-b}{2}}^{\frac{(2j+1)a+b}{2}} e^{iky \sin \theta} dy \right]$$

$0 \rightarrow -N/2$
 $0 \rightarrow N/2$

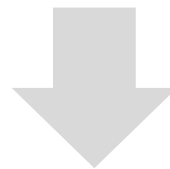
4.6 FRAUNHOFER DIFFRACTION

Many-slit diffraction

$$E(y'; t) = \frac{E_y}{z} e^{ikz - i\omega t} \sum_{j=0}^{j=N/2} \left[\int_{-\frac{(2j+1)a+b}{2}}^{-\frac{(2j+1)a-b}{2}} e^{iky \sin \theta} dy + \int_{\frac{(2j+1)a-b}{2}}^{\frac{(2j+1)a+b}{2}} e^{iky \sin \theta} dy \right]$$

$$\frac{1}{ik \sin \theta} \left[e^{-i\frac{(2j+1)a-b}{2}k \sin \theta} - e^{-i\frac{(2j+1)a+b}{2}k \sin \theta} + e^{i\frac{(2j+1)a-b}{2}k \sin \theta} - e^{i\frac{(2j+1)a+b}{2}k \sin \theta} \right]$$

$$\alpha = \frac{ka \sin \theta}{2}$$



$$\beta = \frac{kb \sin \theta}{2}$$

$$\frac{b}{2i\beta} \left[e^{-i(2j+1)\alpha} (e^{i\beta} - e^{-i\beta}) + e^{i(2j+1)\alpha} (e^{i\beta} - e^{-i\beta}) \right] = b \frac{\sin \beta}{\beta} [e^{-i(2j+1)\alpha} + e^{i(2j+1)\alpha}]$$

$$b \frac{\sin \beta}{\beta} \sum_{j=0}^{j=N/2} [e^{-i(2j+1)\alpha} + e^{i(2j+1)\alpha}] = b \frac{\sin \beta}{\beta} \left(\frac{1 - e^{-iN\alpha}}{1 - e^{-i2\alpha}} e^{-i\alpha} + \frac{1 - e^{iN\alpha}}{1 - e^{i2\alpha}} e^{i\alpha} \right)_{43}$$

4.6 FRAUNHOFER DIFFRACTION

Many-slit diffraction

$$E(y'; t) = \frac{E_y}{z} e^{ikz - i\omega t} b \frac{\sin \beta}{\beta} \left(\frac{1 - e^{-iN\alpha}}{1 - e^{-i2\alpha}} e^{-i\alpha} + \frac{1 - e^{iN\alpha}}{1 - e^{i2\alpha}} e^{i\alpha} \right)$$

$$E(y'; t) = \frac{E_y}{z} e^{ikz - i\omega t} b \frac{\sin \beta}{\beta} \left(\frac{1 - e^{-iN\alpha}}{e^{i\alpha} - e^{-i\alpha}} + \frac{1 - e^{iN\alpha}}{e^{-i\alpha} - e^{i\alpha}} \right)$$

$$= \frac{E_y}{z} e^{ikz - i\omega t} b \frac{\sin \beta}{\beta} \left(\frac{e^{iN\alpha} - e^{-iN\alpha}}{e^{i\alpha} - e^{-i\alpha}} \right)$$

$$\alpha = \frac{ka \sin \theta}{2}$$



$$\beta = \frac{kb \sin \theta}{2}$$

$$I(y') = I_0 \frac{\sin^2 \beta}{\beta^2} \frac{\sin^2 N\alpha}{\sin^2 \alpha}$$

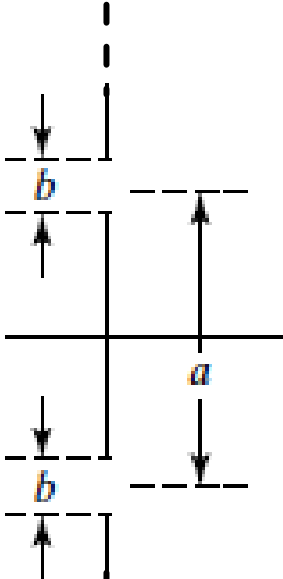
$$E(y'; t) = \frac{E_y}{z} e^{ikz - i\omega t} b \frac{\sin \beta}{\beta} \frac{\sin N\alpha}{\sin \alpha}$$

4.6 FRAUNHOFER DIFFRACTION

Many-slit diffraction

$$I(y') = I_0 \frac{\sin^2 \beta}{\beta^2} \frac{\sin^2 N\alpha}{\sin^2 \alpha}$$

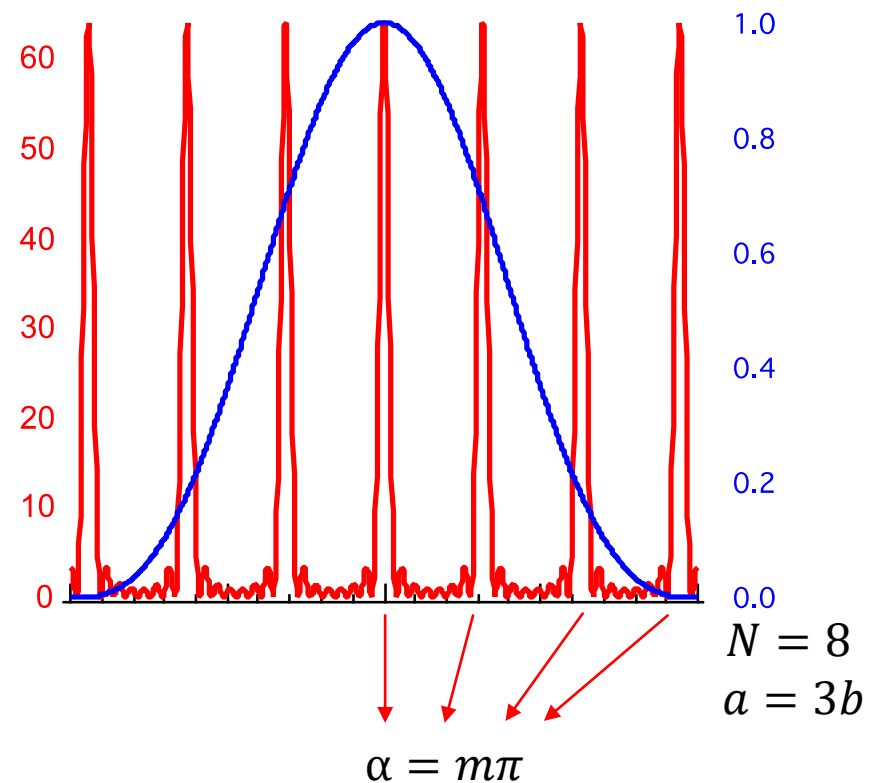
$$\alpha = \frac{ka \sin \theta}{2}$$

$$\beta = \frac{kb \sin \theta}{2}$$


$$\lim_{\alpha \rightarrow m\pi} \frac{\sin N\alpha}{\sin \alpha} = \lim_{\alpha \rightarrow m\pi} \frac{N \cos N\alpha}{\cos \alpha} = \pm N$$

$$\frac{ka \sin \theta}{2} = m\pi$$

$$a \sin \theta = m\lambda$$



m is the order of the diffraction

4.6 FRAUNHOFER DIFFRACTION

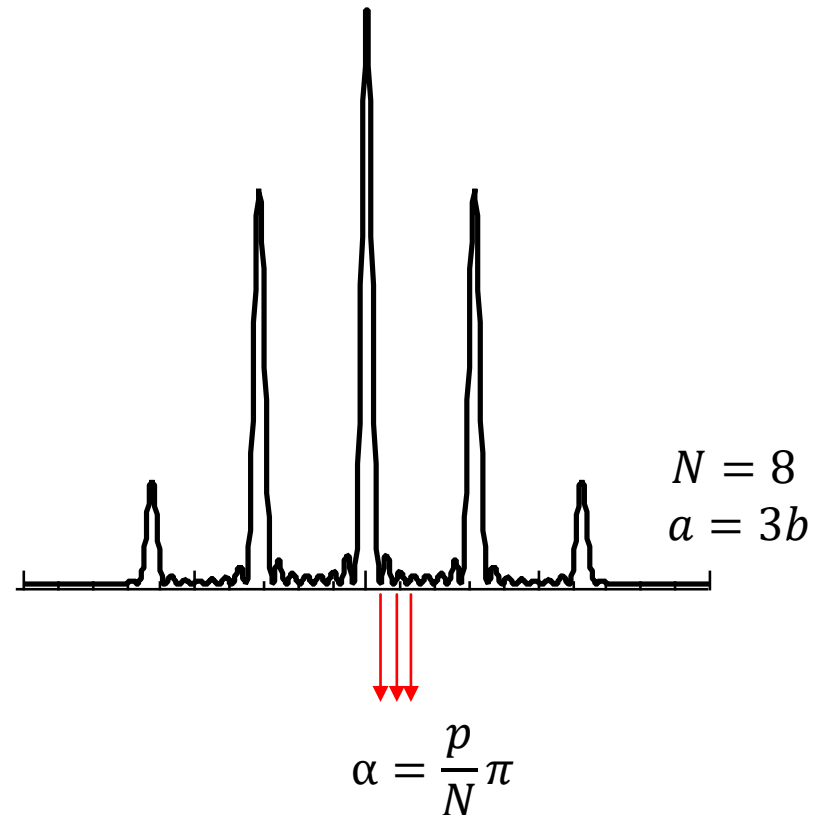
Many-slit diffraction

$$I(y') = I_0 \frac{\sin^2 \beta}{\beta^2} \frac{\sin^2 N\alpha}{\sin^2 \alpha}$$

$$\alpha = \frac{ka \sin \theta}{2}$$

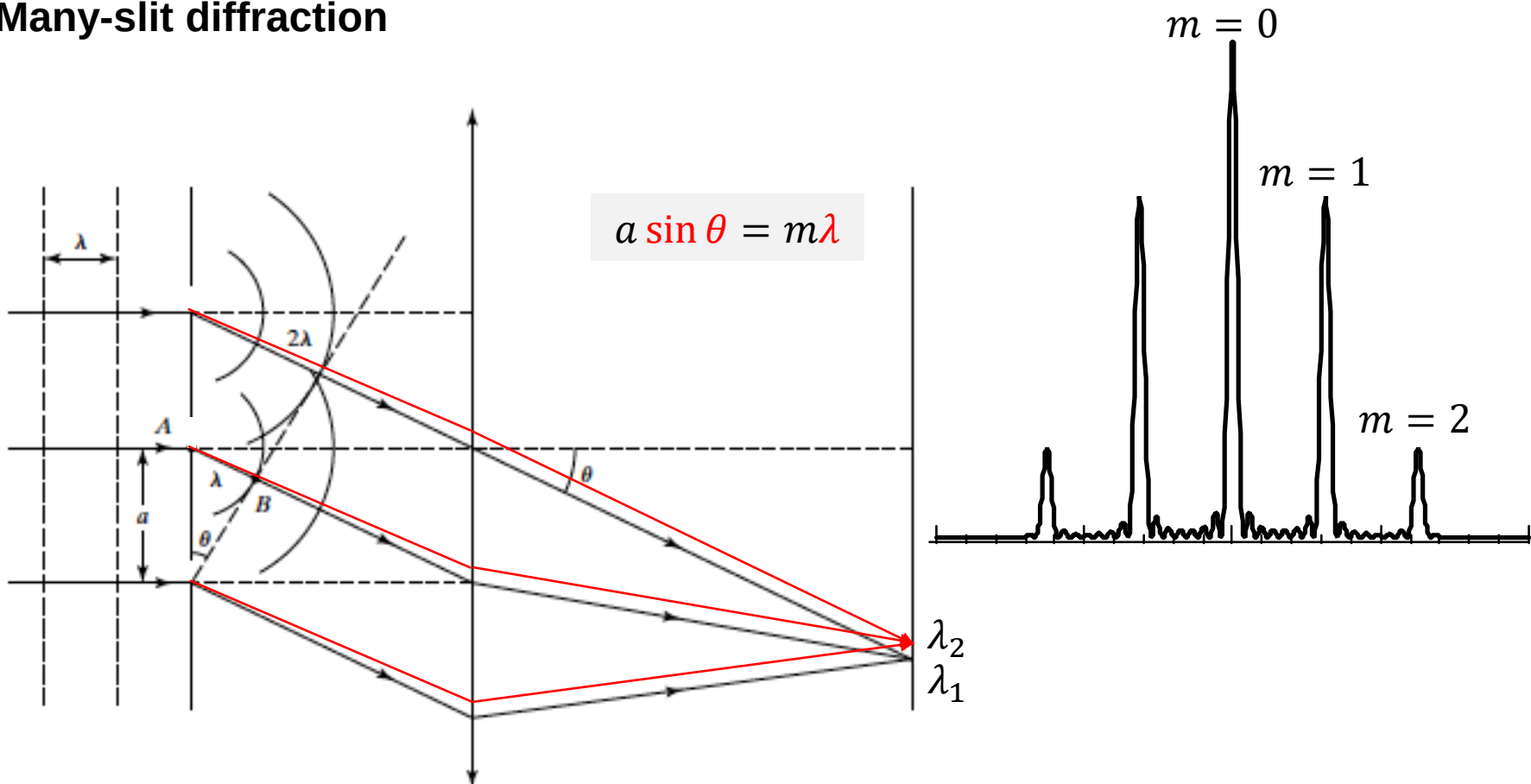
$$\beta = \frac{kb \sin \theta}{2}$$

$$\alpha = \frac{p}{N} \pi \quad \frac{\sin^2 N\alpha}{\sin^2 \alpha} = 0$$



4.6 FRAUNHOFER DIFFRACTION

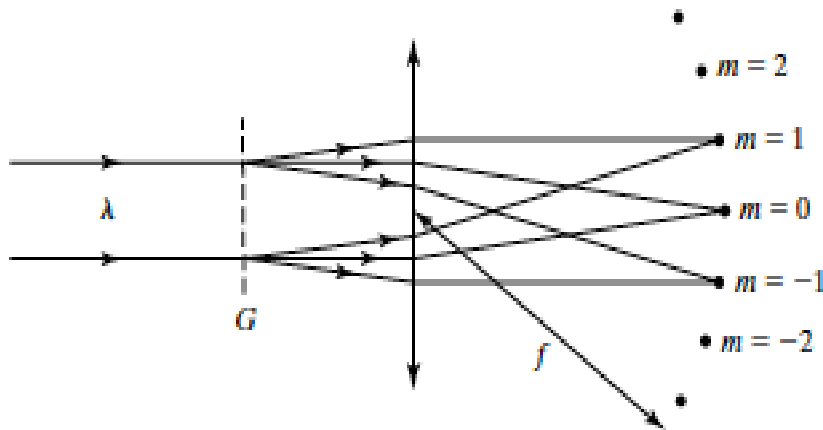
Many-slit diffraction



4.6 FRAUNHOFER DIFFRACTION

Many-slit diffraction

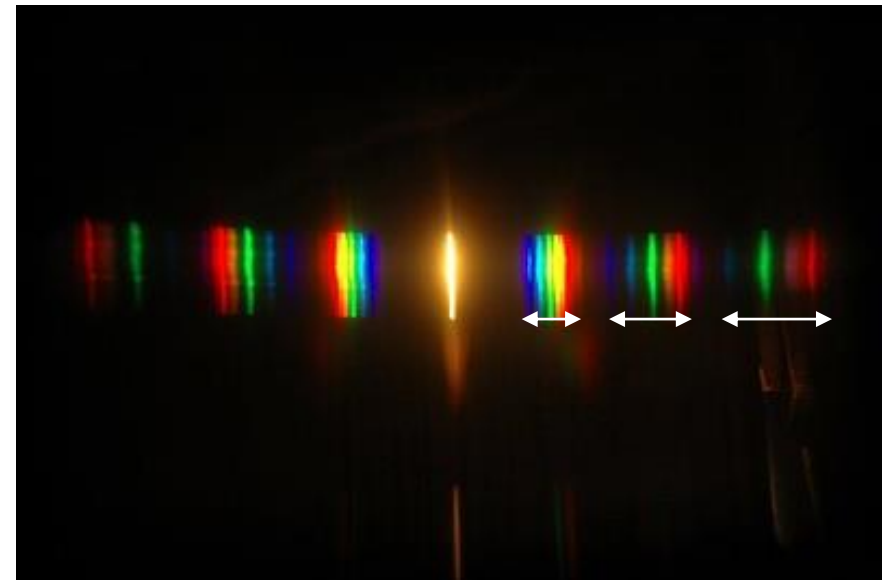
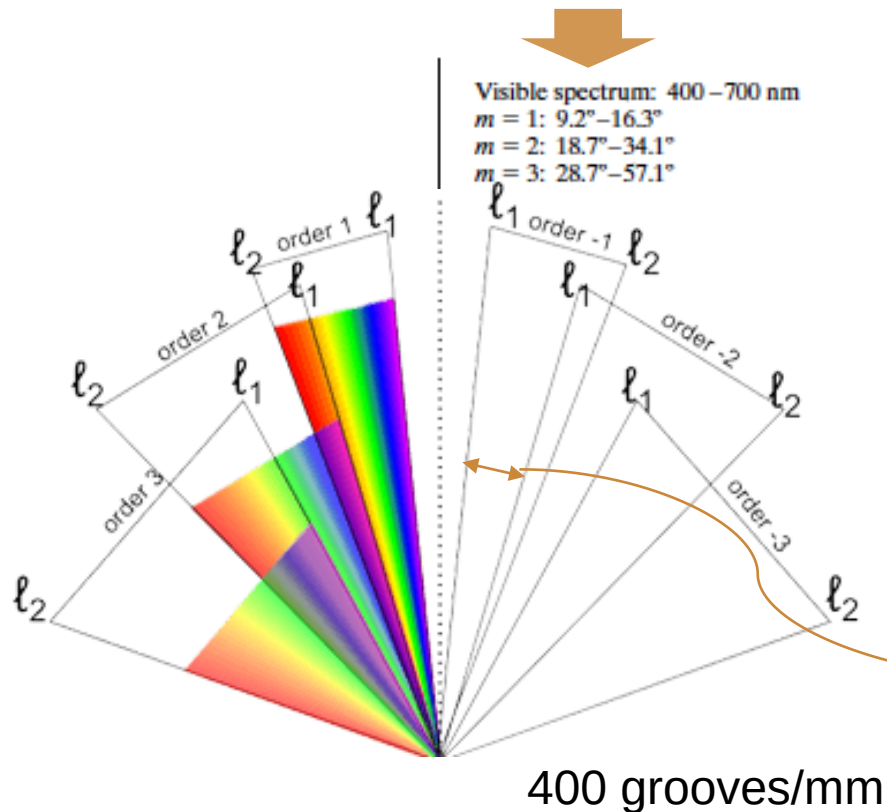
$$a \sin \theta_m = m\lambda$$



4.6 FRAUNHOFER DIFFRACTION

Many-slit diffraction

$$a \sin \theta_m = m\lambda$$



$$m\lambda_2 = (m + 1)\lambda_1$$

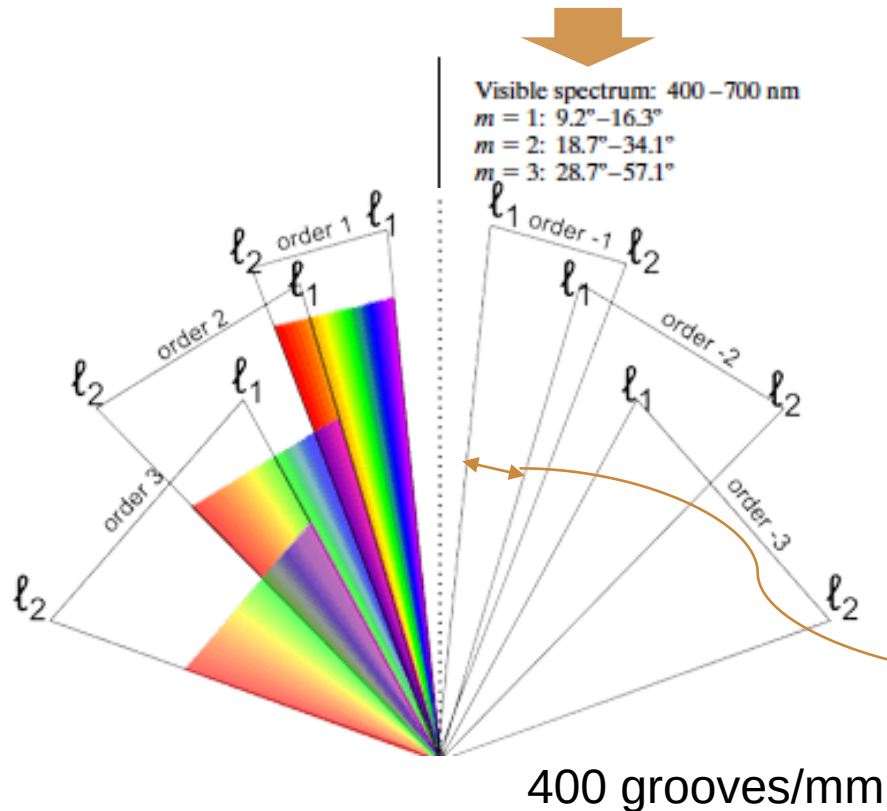
Free spectral range

$$F = \lambda_2 - \lambda_1 = \frac{\lambda_1}{m}$$

4.6 FRAUNHOFER DIFFRACTION

Many-slit diffraction

$$a \sin \theta_m = m\lambda$$



$$a \sin \theta_m = m\lambda$$

Angular dispersion

$$D = \frac{d\theta_m}{d\lambda} = \frac{m}{a \cos \theta_m}$$

$$m\lambda_2 = (m + 1)\lambda_1$$

Free spectral range

$$F = \lambda_2 - \lambda_1 = \frac{\lambda_1}{m}$$

4.6 FRAUNHOFER DIFFRACTION

Many-slit diffraction

Resolving power $R = \frac{\lambda}{\Delta\lambda_{min}}$

the maxima for wavelength $(\lambda + d\lambda)$ is equal to the first minimum for wavelength (λ)

$$m(\lambda + d\lambda) = \left(m + \frac{1}{N}\right)\lambda$$

$$R = \frac{\lambda}{\Delta\lambda_{min}} = mN$$

Rayleigh criterion:

